AI-POWERED SMART SERVICE MARKETPLACE FOR HOUSEHOLD MAINTENANCE IN SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

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Report submitted in partial fulfilment of the requirements for the
BSc (Hons) in Information Technology
Specializing in Information Technology

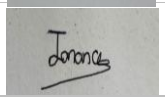
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October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Household maintenance work in Sri Lanka is delivered mainly by independent tradespeople who are found through personal referral. Customers often do not know which trade they need, cannot judge which worker is suitable or trustworthy, and have no reference for a fair price, while capable workers, especially newcomers, remain hard to discover. This report presents an AI-powered smart service marketplace that addresses this common problem with a Flutter mobile application, a Node.js backend, a MongoDB Atlas database and four intelligent cloud microservices. Component 1 maps free-text problem descriptions in English or romanised Sinhala-English (Singlish) to one of 39 worker categories using multilingual sentence embeddings and a support vector machine. Component 2 ranks available workers with a transparent weighted-sum model whose criterion weights were derived by regression from 468 suitability judgements made by 39 hiring professionals. Component 3 fine-tunes XLM-RoBERTa to classify review sentiment in English, Sinhala and Singlish and fuses it with star ratings into a Bayesian-adjusted, category-specific trust score. Component 4 predicts a repair price range with a multi-output decision tree and calculates a fuel-indexed transport fee that responds to live traffic and weather. The components share one 39-label worker vocabulary and exchange data through REST/JSON contracts. All four services run in the cloud, and the team completed the full customer journey in the application, from describing a problem to receiving ranked workers, prices and a post-job trust update. Component results include 78.68% intent accuracy against a 71.50% baseline, ranking of 10,000 candidates in under 5 ms, 95.41% sentiment accuracy and R^2 of 0.888–0.925 for price ranges. These figures are optimistic because much of the data is synthetic or augmented, so validation with real users is the main future work. The project shows that a low-cost, explainable and locally adapted AI marketplace is technically feasible for Sri Lanka.

Keywords: service marketplace, code-mixed Sinhala-English, recommender systems, trust scoring, dynamic pricing

ACKNOWLEDGEMENTS

We express our sincere gratitude to our supervisor, Prof. Dasuni Nawinna, and our co-supervisor, Mr. Nelum Chathuranga Amarasena, of the Faculty of Computing, Sri Lanka Institute of Information Technology, for their guidance, encouragement and constructive feedback throughout this project. Their insistence on empirical justification, honest reporting of limitations and a clear division of work shaped both the integrated solution and the way it is presented in this report.

We thank the forty-one industry professionals, including contractors, site supervisors, service company owners and human resource managers, who completed the worker-suitability survey, and the independent tradespeople and repair workshops who shared their typical labour charges. Their contributions form the empirical foundation of several parts of this work.

We acknowledge the members of public Sri Lankan online community groups whose openly shared household problems helped us understand how people describe repair needs, and the creators of the public datasets and open-source tools on which this work is built. We thank the lecturers and staff of the Department of Information Technology for the knowledge and facilities that made this project possible.

The use of generative AI tools in this project is disclosed in Appendix D. This research received no external funding. Finally, we thank our families and friends for their patience and constant support.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
C1–C4	Component 1 to Component 4
CEYPETCO	Ceylon Petroleum Corporation
CI	Confidence Interval
CPU	Central Processing Unit
FAISS	Facebook AI Similarity Search
GPS	Global Positioning System
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
JWT	JSON Web Token
LKR	Sri Lankan Rupee
MAE	Mean Absolute Error
MCDM	Multi-Criteria Decision Making
NFR	Non-Functional Requirement
NIST	National Institute of Standards and Technology
NLP	Natural Language Processing
NVQ	National Vocational Qualification
OLS	Ordinary Least Squares
PDPA	Personal Data Protection Act
RBF	Radial Basis Function
REST	Representational State Transfer
RMSE	Root Mean Square Error
R^2	Coefficient of determination
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology

Abbreviation	Meaning
SP1–SP4	Subproblem 1 to Subproblem 4
SVM	Support Vector Machine
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
TVEC	Tertiary and Vocational Education Commission
UTF-8	Unicode Transformation Format, 8-bit
WSM	Weighted Sum Model
XLM-R	XLM-RoBERTa, a cross-lingual transformer language model

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Household maintenance in Sri Lanka covers a wide range of everyday work: plumbing, electrical repairs, air-conditioner servicing, carpentry, masonry, painting, appliance repair, cleaning and many specialised trades. Most of this work is carried out by self-employed tradespeople and small workshops rather than by firms with brands or formal listings. The services sector employed 50.6% of the employed population in the fourth quarter of 2025 [1], yet very little of the household repair segment is organised through digital platforms. Customers usually find workers through relatives, neighbours, roadside signboards, classified-advertisement websites and social-media groups, and they agree the price informally, often verbally and on site.

This arrangement works when a trusted contact is available. It breaks down when a household faces an unfamiliar problem, has recently moved, or simply does not know what kind of worker is needed. A customer who sees water dripping from a geyser may not know whether to call a plumber, an electrician or a water-heater technician. Even when the right trade is known, the customer has little evidence about which of several workers is competent, reliable and close enough to come quickly, and no reference for what the job should cost. On the other side of the market, skilled workers without an established referral network, and particularly those who are new to an area, struggle to obtain a steady flow of jobs.

The conditions for a digital intermediary are now in place. At the start of 2025 Sri Lanka had about 12.4 million internet users, an internet penetration of 53.6%, and about 29.3 million cellular mobile connections [2]. Ride-hailing and delivery applications have made app-based booking, rating after service and algorithmic pricing familiar to Sri Lankan users. Internationally, location-based digital labour platforms that match local workers with customers have grown rapidly [3], [4]. However, generic platforms and classified sites assume that customers know the correct category, show ratings that are easy to inflate and hard to interpret, and leave pricing to negotiation. They also operate almost entirely in English, while a large share of everyday written

communication in Sri Lanka uses Sinhala, or Sinhala words written in Latin letters and mixed with English, commonly called Singlish [5], [6].

The research group R26-IT-090 therefore set out to build an AI-powered smart service marketplace for household maintenance in Sri Lanka. The solution is a mobile application for customers and workers, supported by a backend and four intelligent components. Each component addresses one point at which the current informal process fails: finding the right trade, choosing a suitable worker, trusting what other customers say, and agreeing a fair price. This report describes the complete, integrated solution.

1.2 Stakeholders and user needs

The marketplace is two-sided. Its value depends on attracting and retaining both customers and workers, and on giving the platform operator a system that can be explained and defended [7]. Table 1.1 lists the stakeholders identified by the group and the needs that the solution responds to.

Table 1.1: Stakeholders and their needs

Stakeholder	Main needs	How the integrated solution responds
Household customers (primary)	Describe a problem without knowing the trade name; find a suitable, nearby and trustworthy worker quickly; know roughly what the job will cost before committing	Free-text search in English or Singlish (C1); ranked shortlist of suitable workers (C2); text-aware trust scores (C3); repair price range and itemised travel fee (C4)
Workers and technicians (primary)	Receive relevant jobs in their trade; be visible even when new; be judged fairly on reviews; be paid for travel in traffic, rain and periods of high fuel prices	Requests matched to 39 fine-grained trades (C1); cold-start protection and explainable ranking (C2, C3); category-specific reputation (C3); fuel-indexed transport fee passed fully to the worker (C4)
Platform operator (secondary)	Higher booking completion; fewer disputes; ranking and pricing rules that can be audited and adjusted without code changes	Configurable weights, tariffs and scoring constants; decomposable scores; independent cloud microservices
Small workshops and service companies (secondary)	Access to data-driven tools they could not build themselves	Shared platform with recommendation, trust and pricing services

Stakeholder	Main needs	How the integrated solution responds
Regulators and society (indirect)	Protection of personal data; fair and non-misleading prices; inclusion of informal workers in the digital economy	Data minimisation across services; labelled estimates and itemised fees; Sinhala and Singlish support

The needs in Table 1.1 were derived from the group's analysis of the domain, from the existing systems reviewed in Section 2.2, and from direct engagement with stakeholders during the individual components. Component 2 surveyed 41 hiring professionals about how they judge worker suitability. Component 4 collected typical labour charges from tradespeople and repair workshops. Three needs were raised repeatedly: customers want to be understood in their own words, they want a reason to trust a worker they have never met, and both sides want the price to be clear before the visit.

1.3 Alignment with the Sustainable Development Goals

The project aligns most directly with three of the United Nations Sustainable Development Goals (SDGs) [8]. The intended contribution is stated here; Section 7.5 evaluates it critically.

- **SDG 8, Decent Work and Economic Growth.** Target 8.3 encourages the formalisation and growth of micro and small enterprises. The marketplace gives informal tradespeople a digital channel to customers, a portable and verified reputation, compensation for real travel costs, and protection for newcomers through cold-start handling in ranking and trust scoring.
- **SDG 9, Industry, Innovation and Infrastructure.** The project applies modern language technology, recommendation and pricing methods to a traditional, largely offline sector, using low-cost cloud services that a local start-up can operate.
- **SDG 10, Reduced Inequalities.** The system accepts the language people actually use, including romanised Singlish and Sinhala script, rather than requiring formal English trade names. It gives the same transport fee for the same

trip regardless of who the customer is, and reduces the information gap that disadvantages less experienced customers.

Secondary contributions are expected to SDG 11 (Sustainable Cities and Communities), through faster repair of leaks, blocked drains and electrical faults, and SDG 12 (Responsible Consumption and Production), through price information that can favour repair over replacement. These are plausible contributions rather than measured impacts, because the system has not yet been used by the public at scale.

1.4 Purpose and scope of the common report

This common report presents the integrated problem, the complete solution and the evidence that the four components work together as one system. It covers the shared problem definition, the division of that problem into four subproblems, the overall architecture, the interfaces between components, the integrated implementation and deployment, system-level testing, commercialisation and responsible innovation.

The four individual theses provide the detailed research and development evidence for each component, including the full literature reviews, datasets, experimental procedures, detailed designs and component-level results. This report summarises each component only as far as is needed to understand its role and its contribution to the whole, and refers the reader to the relevant thesis for detail:

- Component 1, Intelligent Service Discovery and Intent Recognition, by Bandara W A M S (IT22551320);
- Component 2, Intelligent Worker Recommendation Engine, by Shiwanthika M M (IT22283658);
- Component 3, Automated Review Credibility and Sentiment Analysis, by Wijebandara W.N.S.K (IT22643704);
- Component 4, Dynamic Transport Fee and Repair Price Estimation Engine, by Kumara A.J.T.M (IT22137982).

The remainder of the report is organised as follows. Chapter 2 defines the common problem and its four subproblems and states the group objectives. Chapter 3 presents the overall solution structure and summarises each component. Chapter 4 explains how the components are integrated. Chapter 5 describes the shared implementation and

project execution. Chapter 6 reports integrated testing, results and validation. Chapter 7 addresses commercialisation, sustainability and responsible innovation, and Chapter 8 concludes with future directions.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem addressed by this project is stated as follows:

> In Sri Lanka's informal household maintenance sector, customers cannot reliably find, evaluate, trust and fairly pay a suitable worker for a problem they describe in their own words, and capable workers, particularly new entrants, cannot become visible to customers on fair and transparent terms.

The statement deliberately describes the problem rather than the proposed technology. It contains four failures that occur in sequence during a single customer journey. First, the customer cannot translate an everyday description of a symptom into the correct trade. Second, even with the right trade, the customer cannot tell which of several available workers is most suitable, because the relevant attributes conflict. Third, the evidence about past performance is weak: star ratings are skewed, often based on very few reviews, disagree with what customers write, and the written feedback is multilingual. Fourth, the price is unknown before the visit, and the worker's travel cost in traffic, rain and periods of rising fuel prices is not compensated. A failure at any one of these points ends the journey in the same way: the customer falls back on word of mouth, and the worker loses the job.

The problem matters for three reasons. For customers, delays in fixing leaks, blocked drains and electrical faults have direct costs and safety risks, and the absence of price information exposes them to over-charging. Household repair has the features of a credence good, where the expert both diagnoses the problem and sells the fix, so the customer often cannot verify whether the work or the price was appropriate [9], [10]. For workers, dependence on personal networks concentrates work among those who are already known, and discourages travel to distant jobs or jobs in bad conditions. Fuel shortages and very high inflation after the 2022 economic crisis sharply raised transport costs [11], and fuel prices have continued to change frequently since then [12]. For the digital economy, the lack of a trusted intermediary leaves a large, mostly informal segment of the services sector outside the platforms that have transformed transport and delivery.

2.2 Root causes and overall research gap

2.2.1 Root causes

Five root causes lie behind the four failures.

- **Vocabulary and language mismatch.** Customers describe what they observe ("the sink water is not going down") rather than the trade that fixes it, and much informal writing in Sri Lanka is code-mixed and romanised with no standard spelling. Category menus and English keyword search cannot match such input.
- **Many conflicting selection criteria with unknown importance.** Qualification, reputation, experience and distance all matter, but no single criterion identifies the best worker, and the relative weight that practitioners actually give to each has never been measured for this market.
- **Weak and easily distorted reputation signals.** Rating distributions are strongly skewed towards high values [13], [14]; written text and star scores disagree for a noticeable share of reviews [15]; an average based on one or two reviews is unreliable [16]; and open review systems invite fake reviews [17].
- **Information asymmetry about prices.** Repair prices are rarely published and vary between districts and workers. Without a reference price, customers either accept the first quotation or spend time bargaining, and both outcomes reduce trust [18], [19].
- **Uncompensated and variable travel cost.** A flat call-out charge ignores distance, congestion, weather and fuel prices. Evidence from ride-hailing shows that worker supply responds strongly to price in exactly these conditions [20].

2.2.2 Limitations of current practice and existing systems

Table 2.1 summarises how current alternatives handle the four failures. The comparison draws on the product reviews in the four individual theses.

Table 2.1: How current alternatives address the four failures

Alternative	Finding the trade	Choosing the worker	Trusting reputation	Agreeing the price
Word of mouth and phone calls	Relies on the customer or contact knowing the trade	Personal recommendation only	Informal, unverifiable	Negotiated; no reference price
Classified sites (e.g. ikman.lk)	Category and keyword browsing [21]	Listing order; no suitability ranking	Rarely verified ratings	Occasional advertised starting price
International task platforms (e.g. TaskRabbit)	Customer selects a task category [22]	Rating, completed jobs and responsiveness; weights undisclosed	Ratings, usually from verified bookings; text shown separately	Hourly rates set by workers
Ride-hailing and delivery apps (adjacent market)	Not applicable	Proximity-dominant, opaque	Star ratings, averaged	Algorithmic surge pricing, often seen as opaque [23]
Fixed-menu home service apps	Customer selects from a menu	Assigned by the company	Company-controlled	Fixed company price and visiting charge

Three limitations are common to all of them. They assume that the customer already knows the correct category; none accepts a romanised Singlish problem description. Where they rank workers, the weighting is opaque, and newly registered workers are structurally disadvantaged because the signals that drive ranking are earned through the work that ranking controls access to [24]. And none combines a transparent repair price reference with a travel fee that reflects real travel conditions.

2.2.3 Overall research gap

The research literature addresses parts of the problem in isolation. Work on Sinhala-English code-mixed text has focused on social-media classification tasks such as sentiment and hate speech, on language identification and on transliteration, and existing Sinhala intent classifiers cover small chatbot intent sets [25]. Multi-criteria recommendation methods are well established, but their weights are normally assigned by designers or derived from researcher-supplied pairwise comparisons [26], [27]. Reputation systems and Bayesian rating adjustments are mature but operate on star ratings and ignore review text [28], [29], [30]. Dynamic pricing research has

concentrated on demand-based surge pricing for ride-hailing rather than cost-based travel fees for household technicians [20], and machine-learning price estimation has mostly predicted a single value from rich numerical features in developed markets.

The overall gap is therefore the absence of an integrated, locally adapted system for Sri Lankan household services that (a) understands problem descriptions in English and romanised Singlish, (b) ranks workers with criterion weights derived from practitioner judgement, (c) builds a text-aware, sample-size-aware and category-specific reputation from verified multilingual reviews, and (d) gives both sides a transparent price reference for the repair and the travel. Each of (a) to (d) is a research gap in its own right, documented in the individual theses. The integrated gap is that they have not been addressed together, on a shared vocabulary of trades, in one working application.

2.3 Subproblem and component mapping

The common problem was divided into four subproblems, one per team member. Figure 2.1 shows the decomposition and Table 2.2 states each subproblem, its boundary and the reason it is necessary.

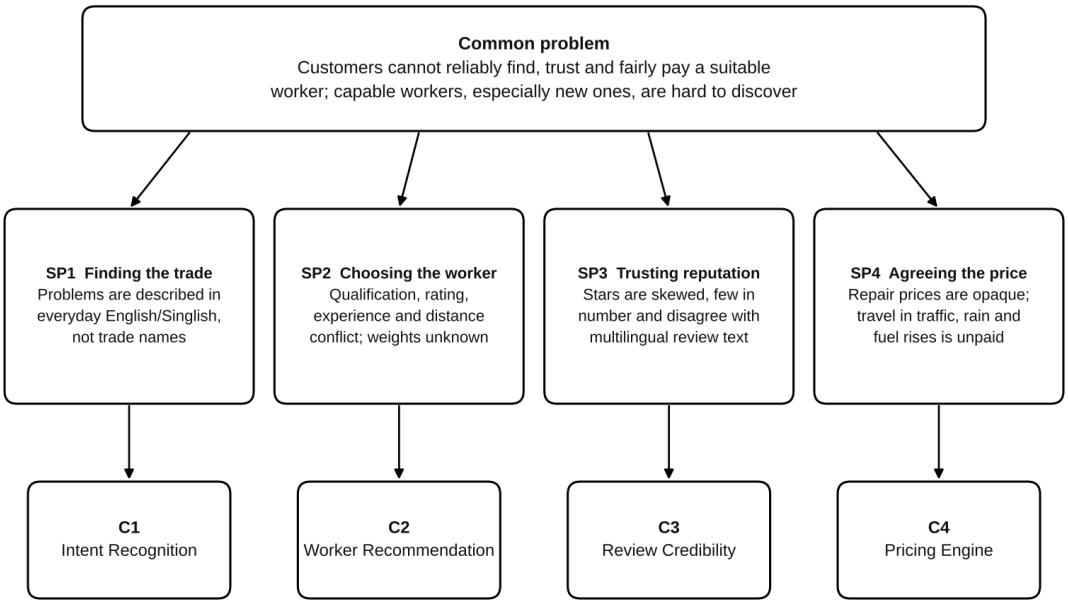


Figure 2.1: Decomposition of the common problem into four subproblems and components

Table 2.2: Subproblems, component boundaries and necessity

Subproblem	Definition	Boundary (what it does not do)	Why the solution fails without it
SP1 Finding the trade (C1)	Map one free-text household problem description in English or romanised Singlish to one of 39 worker categories, and decline when unsure	Does not rank workers, judge quality or estimate price; does not predict a sub-issue or handle multi-trade requests	No customer can start the journey without knowing the trade name; every later step depends on the category
SP2 Choosing the worker (C2)	Rank available workers in the requested category by suitability, using empirically justified weights, and keep new workers discoverable	Does not resolve categories, analyse review text or price jobs	The customer receives an unordered list and must judge conflicting attributes alone
SP3 Trusting reputation (C3)	Turn verified stars and multilingual review text into a credible, category-specific trust score	Does not detect fake reviews with machine learning (prevented by verified completion); does not rank workers	Reputation stays a skewed star average that ignores what customers write and overreacts to a single review
SP4 Agreeing the price (C4)	Give the customer a repair price range and the technician a context-aware transport fee	Does not handle payment, commission or spare-part pricing; does not use trust scores in pricing	Customers cannot form fair expectations, and technicians avoid distant jobs or jobs in bad conditions

The subproblems are non-overlapping: each component owns one decision, and no decision is made by two components. They are, however, interdependent. C1's category is the filter for C2, the category under which C3 stores reputation, and the worker type for which C4 predicts a price. C2's ranking depends on reputation data produced after jobs are reviewed, which C3 analyses. C4's transport fee needs the location of the technician the customer selects from C2's list. The shared vocabulary of 39 worker categories, described in Section 4.2, is what allows the four decisions to be made consistently.

2.4 Overall aim and group objectives

The overall aim of the project is to design, implement, integrate and evaluate an AI-powered mobile marketplace for household maintenance in Sri Lanka that lets a customer describe a problem in their own words and then find, trust and fairly pay a suitable worker, while giving workers fair visibility and compensation.

The group objectives are listed in Table 2.3. Each objective is mapped to the components that deliver it and to the evidence used to evaluate it in Chapter 6.

Table 2.3: Group objectives, responsible components and evaluation evidence

ID	Objective	Components	Evaluation evidence
GO1	Accept free-text problem descriptions in English and romanised Singlish and map them to one of 39 worker categories, with accuracy and macro-F1 clearly above a similarity-search baseline, and decline low-confidence queries	C1	Held-out accuracy and macro-F1 against FAISS baseline; rejection rule behaviour
GO2	Rank available workers with empirically derived criterion weights, producing scores that are bounded, monotonic, deterministic and explainable, returning a top-20 list within 2 seconds and keeping new workers discoverable	C2	Regression weights and significance; functional property tests; latency by pool size; cold-start effect
GO3	Produce a category-specific trust score from verified reviews in English, Sinhala and Singlish, with sentiment accuracy of at least 85%, that does not overreact to few reviews	C3	Held-out accuracy and F1; scoring scenarios
GO4	Provide an explainable repair price range for a job and a transparent, context-aware transport fee for the technician, both delivered as low-latency services	C4	R^2 and MAE against a random forest baseline; fee scenario analysis; automated tests; latency
GO5	Integrate the four components into one working mobile marketplace in the cloud, using a shared category vocabulary and documented API contracts, so that the complete customer journey runs end to end	All	End-to-end test of the customer journey; interface tests; deployment status
GO6	Validate the solution with stakeholders and assess its commercial viability, responsible-AI risks and SDG contribution	All	Practitioner survey; commercialisation and compliance analysis

2.5 Scope assumptions and constraints

In scope. The solution covers the customer journey from a free-text search to a completed and reviewed job: intent recognition, worker ranking, price estimation, transport fee calculation, booking, and review-based trust scoring. It supports English and romanised Singlish for problem descriptions, and English, Sinhala script and

Singlish for reviews. It covers the 39 worker categories listed in Appendix C. The first target market is urban and suburban households in the Western Province.

Out of scope. The following are excluded from the current solution: payment processing and commission handling (the platform rule that the full transport fee goes to the technician is applied in the booking flow, not computed by the AI components); Tamil script and Tamil-English input; multi-trade requests and sub-issue prediction; machine-learning detection of fake reviews (prevented instead by accepting reviews only for completed jobs); spare-part pricing; and personalised ranking per customer.

Assumptions. The main assumptions are that a customer's request concerns one trade; that a review can be written only after the job is marked completed in the application; that the worker attributes used for ranking (qualification, star rating, experience, location and availability) are recorded in the platform database; that Google Maps traffic durations are a reasonable proxy for travel effort; and that the surveyed price ranges and practitioner judgements represent typical urban conditions in Sri Lanka in 2026.

Constraints. The project was carried out by four undergraduate students within one academic year, with no external funding. The platform had not been launched, so no historical booking, review or price data existed. This constraint shaped several design choices: training data had to be generated, augmented or surveyed; models had to train on free cloud notebooks; and services had to run on low-cost cloud tiers. Free-tier hosting can add a delay of up to about ninety seconds when an idle service receives its first request.

Dependencies. The transport fee depends on the Google Maps Distance Matrix API and the OpenWeatherMap API [31], [32]. All components depend on the shared 39-label category vocabulary and on the MongoDB Atlas database maintained by the backend.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The solution is one product: a mobile marketplace in which a customer describes a household problem in their own words and leaves the application with a suitable, trusted worker booked at a price that both sides understand. The four intelligent components are not separate applications. Each one makes a single decision at a specific point in the customer journey, and each decision feeds the next.

The central value proposition can be stated in one sentence: **"Describe your problem the way you would tell a neighbour, and get the right worker, a reason to trust them, and a fair price, before anyone comes to your door."** The end-to-end outcome for the customer is a completed job with no need to know trade names, compare workers manually or bargain on site. For the worker, the outcome is a stream of jobs in their own trade, visibility from the first day on the platform, a reputation that reflects what customers actually wrote, and payment for the real cost of reaching the customer.

Three design principles run through all four components and give the solution its coherence:

- **Explainability over opacity.** Every score or price that a customer or worker sees can be decomposed. C1 returns a confidence and declines when unsure; C2 returns the contribution of each criterion to a worker's score; C3 stores the star score, sentiment score and credibility factor behind each trust score; C4 returns an itemised transport fee.
- **Fairness to newcomers.** Both the ranking and the reputation logic contain explicit cold-start handling, so that a worker with no history is treated as unproven rather than as poor.
- **Local adaptation at low cost.** The components support Singlish and Sinhala, use a Sri Lankan qualification framework and Sri Lankan prices and tariffs, and run on CPU-based, low-cost cloud services.

3.2 High level architecture

Figure 3.1 shows the architecture of the complete solution. It has four tiers.

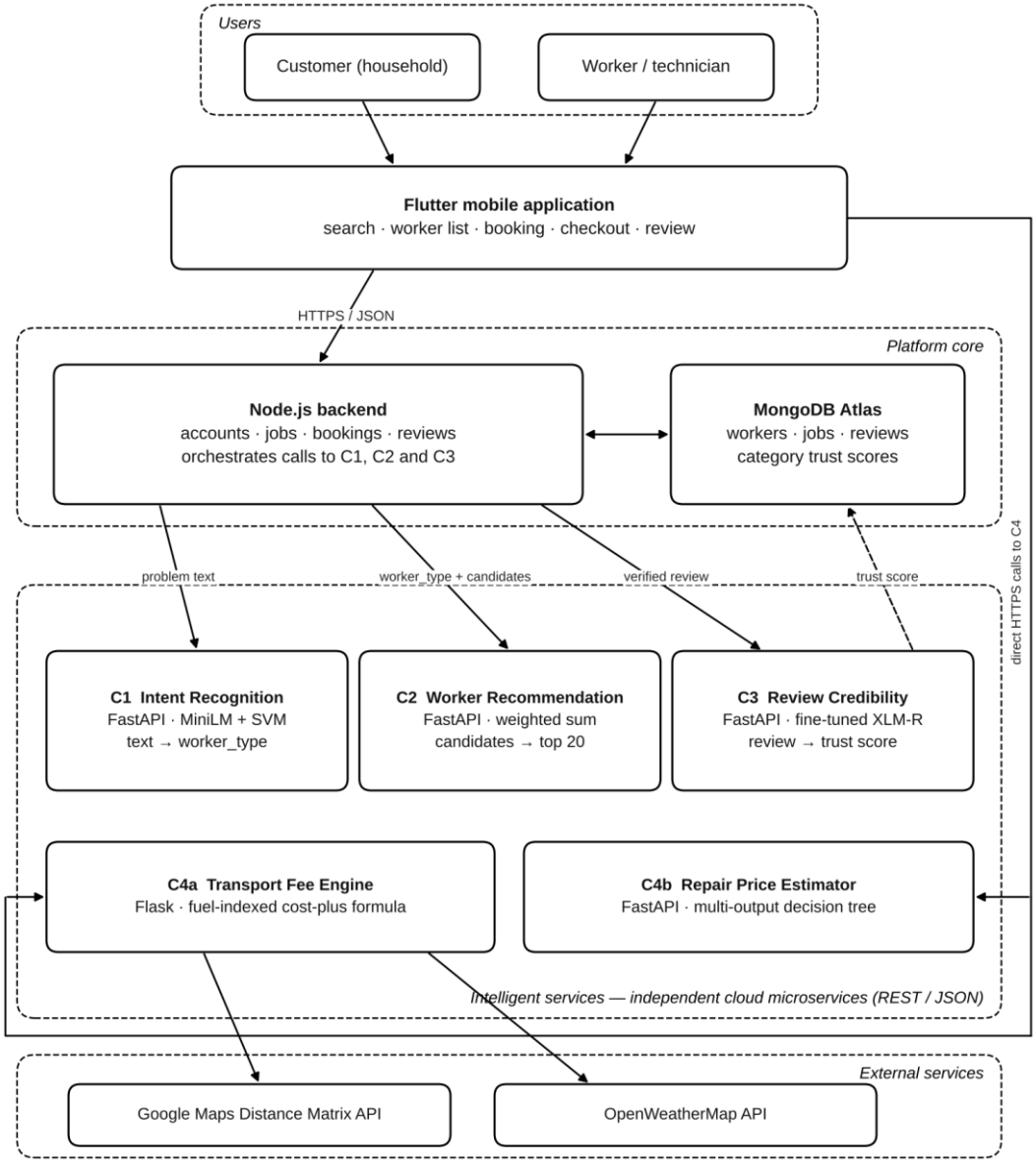


Figure 3.1: High-level architecture of the integrated marketplace

Client tier. A Flutter mobile application [33] serves customers and workers. It provides the free-text search bar, the ranked worker list, the repair price display, checkout with the transport fee breakdown, and the post-job review screen.

Platform core. A Node.js backend holds the business logic for user accounts, worker profiles, jobs and their status, bookings and reviews. It stores data in a managed MongoDB Atlas database [34]. The backend orchestrates the calls to Components 1,

2 and 3: it forwards the search text to C1, assembles the candidate worker records for C2, and passes verified reviews to C3.

Intelligent services. The four components run as independent Python microservices in the cloud, each exposing a REST interface with JSON payloads [35]. Components 1, 2 and 3 and the repair price estimator of Component 4 use FastAPI [36]; the transport fee engine of Component 4 uses Flask. Each service loads its model or configuration once at start-up and handles each request statelessly. Component 4's two modules are currently called directly by the Flutter application; routing them through the backend gateway is planned (Section 4.1).

External services. The transport fee engine calls the Google Maps Distance Matrix API for road distance and traffic duration, and the OpenWeatherMap API for the weather at the customer's location [31], [32].

The microservice split was chosen jointly by the group for four reasons [37], [38]. The machine-learning stacks are in Python while the backend is in Node.js. Each team member could develop, test and deploy their component independently. A failure in one service, such as an external weather API timing out, does not stop the others. And each service can be scaled according to its own load: the sentiment model is the heaviest to run, while the ranking and pricing services need only milliseconds of computation per request.

3.3 Component 1 summary

Table 3.1: Component 1 at a glance

Item	Description
Title and owner	Intelligent Service Discovery and Intent Recognition; Bandara W A M S (IT22551320)
Subproblem	SP1: customers describe symptoms in English or romanised Singlish rather than naming the trade
Input	One free-text problem description, e.g. "watura bata pupurala" (the water pipe has burst)
Core method	Raw text encoded without pre-processing by the pretrained multilingual sentence-embedding model paraphrase-multilingual-MiniLM-L12-v2 (384 dimensions, not fine-tuned); RBF-kernel support vector machine classifier with probability estimates; rejection rule at 0.4 confidence

Item	Description
Output	{worker_type, confidence}, where worker_type is one of 39 lower-case labels, or a "not found" result below 0.4
Key result	78.68% accuracy and 0.79 macro-F1 on a 4,259-row held-out test set, against 71.50% and 0.71 for FAISS nearest-neighbour search; 78.35% and 0.78 on a cleaned dataset
Contribution to the whole	The entry point of the journey and the source of the shared category label used by C2, C3 and C4

Component 1 replaces the category menu found in existing platforms with free-text search. Customers can write as they would speak to a neighbour, for example "current eka wada na" (the electricity is not working) or "AC eka sisil wenne na" (the air conditioner is not cooling). The component returns a single worker category from a taxonomy of 39 trades, ranging from AC technician to well cleaner.

A balanced dataset of 21,294 descriptions (546 per category, about 42% romanised Singlish) was built from team-written seed phrases, problems re-written from public Facebook groups, and AI-assisted generation. Rather than relying on hand-built Singlish spelling rules, which do not exist as a public resource, the component relies on the sub-word tokenizer of a multilingual sentence encoder [39]. Two decision methods were compared on identical embeddings and the same stratified 80/20 split. The SVM [40] improved F1 in 36 of the 39 categories, and the 95% confidence intervals for the two accuracies do not overlap. Remaining errors cluster in closely related trades such as carpenter and furniture restoration, and the bathroom and kitchen water-heater categories.

Two design choices matter for the integrated system. First, the output is a single flat label, which keeps the contract with the other components narrow and stable. Second, the 0.4 rejection threshold keeps the application from confidently sending a customer to an unrelated trade; for an ambiguous query such as "lin sudda karanawa" (cleaning the well) the top probability was only 0.17 and the query was declined. The main limitation is that most of the data is AI-generated, so the accuracy describes performance on synthetic descriptions, and validation on real customer queries is the main next step.

3.4 Component 2 summary

Table 3.2: Component 2 at a glance

Item	Description
Title and owner	Intelligent Worker Recommendation Engine; Shiwanthika M M (IT22283658)
Subproblem	SP2: several conflicting criteria determine suitability, and their relative importance is unknown
Input	worker_type from C1 (applied as the category filter), the customer's location, and candidate worker records with NVQ level, star rating and review count, years of experience, distance and availability
Core method	Four-stage pipeline: eligibility filtering (category, online status), min–max normalisation with distance inverted to proximity, weighted sum with survey-derived weights, and ranking; neutral rating of 3.5 for workers with fewer than three reviews
Output	Top 20 workers by suitability score (0–100), with an optional per-criterion breakdown
Key result	Weights derived from 468 judgements by 39 hiring professionals: proximity 0.396, customer rating 0.305, experience 0.215, NVQ level 0.084; 10,000 candidates ranked in 4.62 ms at the 95th percentile
Contribution to the whole	Turns a category into an ordered, explainable shortlist and keeps new workers visible

Component 2 decides which workers a customer sees and in what order. This decision has a direct economic effect on workers, so the component was designed to be auditable as well as accurate. It uses the Weighted Sum Model, the most transparent multi-criteria aggregation method [41], because every score can be decomposed exactly into the contributions of its criteria. A supervised model trained on booking outcomes was rejected because the platform has no outcome history before launch, and TOPSIS [42] was set aside because it still needs external weights and its scores are less directly interpretable.

The research contribution lies in how the weights were obtained. An earlier iteration derived weights by the Analytic Hierarchy Process from literature-based pairwise comparisons [27], but those judgements were ultimately supplied by the researcher. They were replaced with a profile-based survey: 41 hiring professionals rated twelve worker profiles in which the four criteria were varied almost independently (maximum correlation 0.208). After removing two non-discriminating responses, ordinary least squares regression on 468 standardised observations gave the weights in Table 3.2 (R^2

= 0.200). Three coefficients were significant at $p < 0.001$; the NVQ coefficient was not ($p = 0.092$) and was retained on policy grounds, because the Tertiary and Vocational Education Commission's Recognition of Prior Learning pathway makes qualification a meaningful competence signal [43].

A secondary finding is a clear gap between stated and revealed preference [44]. When the same professionals ranked the criteria directly, they placed experience first and distance last, while their actual evaluations weighted proximity most heavily (Spearman correlation −0.400, stable across four rank-to-weight conversion methods). The implementation produces scores bounded between 0 and 100 that are monotonic in all four criteria, identical across repeated runs, and decomposable. The cold-start rule adds 19.04 points to a new worker with a strong qualification compared with treating the missing rating as zero.

3.5 Component 3 summary

Table 3.3: Component 3 at a glance

Item	Description
Title and owner	Automated Review Credibility and Sentiment Analysis; Wijebandara W.N.S.K (IT22643704)
Subproblem	SP3: star ratings are skewed, unreliable when few, disagree with review text, and the text is multilingual
Input	A review for a completed job: worker, service category (the job's worker_type), star rating (1–5) and optional text in English, Sinhala script or Singlish
Core method	Fine-tuned xlm-roberta-base classifier (Negative, Neutral, Positive); sentiment value = polarity × confidence; fusion of 60% star score and 40% sentiment score on a 0–10 scale; Bayesian adjustment Final = C × Raw + (1 − C) × 6.0 with credibility C = N/(N + 5); scores kept separate per category
Output	Category-specific trust score (0–10) with the review count and credibility factor, stored in MongoDB Atlas and available through the service API
Key result	95.41% accuracy and 0.954 weighted F1 on 16,853 held-out rows; scoring behaves as designed in eight verification scenarios
Contribution to the whole	Makes reputation credible: verified reviews only, text-aware, robust to a single review and isolated by trade

Component 3 is triggered when a customer leaves feedback on a completed job. Fake reviews are prevented by a business rule rather than by a classifier: only a customer

whose job has been marked completed can write a review [17]. In this component, credibility means how much confidence the system can place in a score given the number of verified reviews behind it.

A three-class sentiment dataset of 101,217 rows was built by extending a public English Twitter dataset with Sinhala machine translations, rule-based Sinhala-to-Singlish transliterations and template-generated Singlish sentences. After removing the Irrelevant class, 84,264 rows remained. The multilingual transformer xlm-roberta-base [45] was fine-tuned on 67,411 rows. It classified all positive and negative smoke-test sentences correctly in all three language forms, including informal Singlish such as "wadak na ade mu", but misclassified two of three neutral sentences and showed overconfidence after the fifth epoch.

The scoring model addresses the three weaknesses of star-only ratings. Fusing stars with text sentiment lets a clear complaint lower a high star rating: in one scenario, near-perfect stars with negative text gave a raw score of 6.68 instead of 9.80. The Bayesian adjustment [30] gives a new worker a neutral starting score of 6.0 and limits the effect of a single review: one perfect review raises the score only to 6.66, while an established worker with fifty excellent reviews reaches 9.15. Category isolation means that a poor electrical job does not reduce a worker's plumbing reputation. Because the training data is augmented and comes from tweets rather than service reviews, the reported accuracy is likely to be optimistic.

3.6 Component 4 summary

Table 3.4: Component 4 at a glance

Item	Description
Title and owner	Dynamic Transport Fee and Repair Price Estimation Engine; Kumara A.J.T.M (IT22137982)
Subproblem	SP4: customers cannot judge a fair repair price, and technicians are not compensated for travel in traffic, rain and periods of high fuel prices
Input	Repair estimate: worker_type (from C1), service and specification. Transport fee: the selected technician's GPS location (from C2's list) and the customer's location

Item	Description
Core method	Repair Price Range Estimator (FastAPI): one-hot encoded multi-output decision tree predicting minimum and maximum price. Transport Fee Engine (Flask): fee = [LKR 200 + distance × LKR 80/km × (current petrol price / LKR 350)] × traffic multiplier (1.00, 1.15, 1.30) × weather multiplier (1.00, 1.20, 1.35)
Output	{price_min, price_max} in LKR; itemised transport fee (base fare, distance cost, traffic and weather premiums)
Key result	R^2 0.888 (minimum) and 0.925 (maximum), MAE LKR 1,712 and 3,144, comparable to a 100-tree random forest; fee bounded at 1.755 times the normal fee; 14/14 automated tests; mean estimator latency 11.10 ms
Contribution to the whole	Closes the journey with transparent prices for both sides of the market

Component 4 handles the money side of a booking and has two decoupled modules. The **Repair Price Range Estimator** shows the customer an indicative price range as soon as a service is selected, before any technician is chosen. It was trained on 630 rows derived from 105 surveyed Sri Lankan price combinations covering 39 trades, 95 services and 91 specifications, using the same 39 worker categories as C1. A single decision tree [46] was preferred to a random forest because it achieved comparable accuracy with lower MAE on both targets, can be read as rules, and always predicts a minimum no greater than the maximum.

The **Context-Aware Dynamic Transport Fee Engine** calculates the technician's travel fee at checkout. It deliberately uses a transparent cost-plus formula rather than demand-based surge pricing, because cost-justified price changes are perceived as fair [47], [19] while surge pricing is widely criticised as opaque [23]. The distance rate is indexed to the CEYPETCO petrol price, so the fee follows fuel price revisions without any code change: across the 2026 revisions, the fee for a 10 km trip changed by 7.4%. For a 5.4 km trip in heavy traffic and rain, the fee is LKR 1,109, of which roughly a third is due to the conditions at that moment, and the application shows this breakdown. The whole transport fee is passed to the technician; platform commission applies only to repair labour. The repair results are optimistic because augmented copies of the same base combination could fall in both training and test sets.

3.7 End to end user and data flow

Figure 3.2 shows how a single customer request moves through the complete solution and where each component acts.

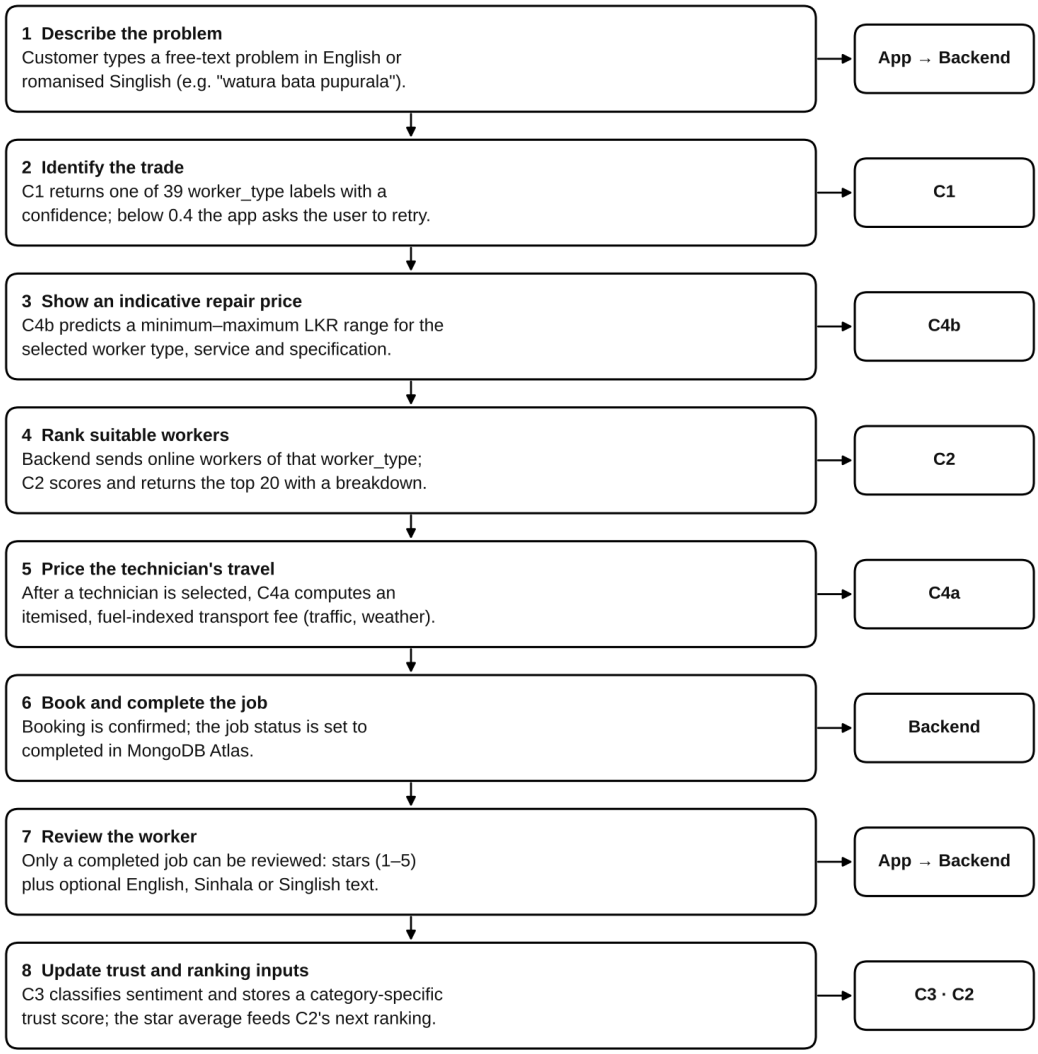


Figure 3.2: End-to-end customer journey and the component responsible for each step

The journey has eight steps.

- **Steps 1 and 2: search.** The customer types the problem in the search bar. The backend forwards the text to C1, which returns a worker_type and a confidence. If the confidence is below 0.4, the application shows "We couldn't find the worker, try again" and the customer can rephrase.

- **Step 3: repair estimate.** For the identified worker type, the customer selects a service and specification, and the application requests a price range from C4's estimator. The range is labelled as an estimate.
- **Step 4: recommendation.** The backend queries MongoDB Atlas for workers whose category matches the worker_type and assembles their records, including each worker's star average and review count. C2 removes offline workers, scores the rest and returns the top twenty, which the application displays.
- **Step 5: checkout.** When the customer selects a worker, the application sends that worker's current GPS position and the customer's location to C4's transport fee engine, which returns an itemised fee.
- **Step 6: booking and completion.** The customer confirms the booking. When the job is finished, its status is set to completed in the database.
- **Steps 7 and 8: review and update.** The review screen is available only for completed jobs. The customer gives a star rating and, optionally, a written review. The backend checks the job status and passes the review to C3, which classifies any text, stores the review under the worker and the job's category, recomputes the category trust score, and saves it. The updated star average is used in the worker's next ranking by C2.

The flow closes a loop: every completed job produces reputation data that changes future rankings. The same category label created in step 2 is used for the price estimate in step 3, the candidate filter in step 4 and the reputation record in step 8. The way the data moves between the components, and what happens when something fails, are described in Chapter 4.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The group followed an **interface-first, independently deployable** integration strategy. Integration risk in a four-person project comes mainly from mismatched assumptions between components, so the team agreed the contracts between components early and kept them as narrow as possible. Each component was then developed, tested and deployed by its owner as a separate service, and integrated into the application through its agreed interface.

Integration proceeded in four stages.

- **Stage 1: shared vocabulary and contracts.** The first decision was the category taxonomy. C1 defined 39 worker categories and their labels, and the other three components adopted the same set: C2 uses it as the category filter on worker records, C3 stores reputation under it, and C4 built its repair price dataset on it. The group then agreed one message per interaction (Section 4.2).
- **Stage 2: independent services.** Each component was exposed as a stateless REST service that could be tested in isolation with synthetic inputs. C2, for example, deliberately holds no database connection: the caller supplies the candidate worker records in the request, so the ranking logic could be tested with thousands of synthetic workers before the database existed.
- **Stage 3: application integration.** The backend and the Flutter application were connected to each service in the order of the customer journey: C1 to C2 (search and recommendation), then C4 (prices at service selection and checkout), then C3 (reviews after completion). Early integration ran on a local development machine.
- **Stage 4: cloud deployment and end-to-end testing.** All four services were moved to the cloud and connected to the application through their HTTPS endpoints. The team then ran the complete customer journey in the application (Section 6.2).

Interface ownership. The owner of each service owns its interface definition and is responsible for validating its input and for documenting its responses. The backend

team, shared by all members, owns the orchestration logic and the database schema. Table 4.1 lists the owners.

Table 4.1: Interface ownership

Interface	Owner	Consumer
Search text to worker category	C1 (Bandara W A M S)	Node.js backend
Candidate workers to ranked shortlist	C2 (Shiwanthika M M)	Node.js backend
Review to sentiment and trust score	C3 (Wijebandara W.N.S.K)	Node.js backend; MongoDB Atlas
Job to repair price range	C4 (Kumara A.J.T.M)	Flutter application
Trip to transport fee	C4 (Kumara A.J.T.M)	Flutter application
Category label set (39 labels)	C1, agreed by all	C2, C3, C4, backend, application

Integration risks and how they were managed. Four risks were identified. *Label mismatch*, where a correct C1 prediction matches no workers because another component spells the category differently, was managed by adopting one label list as the shared reference (Appendix C). *Scale mismatch* between the reputation produced by C3 (0 to 10) and the rating expected by C2 (1 to 5) was managed by staging the integration, as explained in Section 4.2.3. *External dependency failure* in C4 was managed with timeouts and neutral fallbacks (Section 4.4). *Hosting latency*, where a free-tier service takes up to about ninety seconds to wake after inactivity, was mitigated by issuing a health request when the application starts, so that services begin resuming before the user needs them.

Planned gateway routing. C4's endpoints are currently called directly from the Flutter application, which allowed C4 to be developed and tested independently. In the target production architecture, these calls will be routed through the Node.js backend acting as an API gateway, which will add JSON Web Token authentication, rate limiting and booking metadata [48].

4.2 Interfaces data contracts and shared models

4.2.1 Shared identifiers and the category vocabulary

Three identifiers are shared across the system: the **worker category**, the **worker identifier** and the **job identifier**. The worker category is the most important. It is created by C1 and reused, without translation, by every other part of the system. C1 emits the category in a machine form (lower-case with underscores, for example ac_technician). The same 39 categories appear in a readable form (for example "AC Technician") in the user interface and in C4's repair price dataset. Appendix C lists both forms. Any new category must be added to C1's training data, to the worker registration options, to C4's dataset and to the label list at the same time.

4.2.2 Interface catalogue

Table 4.2 summarises the interfaces between the application, the backend and the four services. The full request and response formats are given in Appendix A.

Table 4.2: Interface catalogue of the integrated solution

Service	Endpoint	Request	Response
C1 Intent Recognition	POST prediction endpoint	{problem_description} (English or Singlish text)	{worker_type, confidence}; or {worker_type: null, confidence, message} below 0.4
C2 Worker Recommendation	POST /recommend (?explain=true); GET /weights; GET /health; POST /score	{category, worker_type (optional), top_n, workers[ ]} with each worker's id, nvq, rating, review_count, experience, distance_km, availability	{category, total_received, eligible_after_filter, returned, results[ ]} with rank, id, final_score, cold_start_applied and optional breakdown
C3 Review Credibility	Analyse review; get trust score	Review: {worker, category, stars, text (optional)}. Score: {worker, category}	Review: {label, confidence, sentiment value, updated category score}. Score: {trust_score (0–10), review_count, credibility}
C4 Repair Estimator	POST /api/estimate-repair	{trade, task, sizing}	{trade, task, sizing, price_min, price_max, currency: "LKR"}
C4 Transport Fee	POST /api/estimate-transport	{origin: {lat, lng}, destination: {lat, lng}}	{distanceKm, trafficDelayMins, weatherCondition, totalTransportFee, breakdown{baseFare, distanceCost, trafficMultiplier, weatherMultiplier, fuelRatio}}

All services exchange UTF-8 JSON over HTTPS, which is required for Sinhala script in reviews. The FastAPI services generate OpenAPI documentation automatically from their schemas, which gives the other team members a precise, always-current contract.

4.2.3 The reputation interface between C3 and C2

The connection between Component 3 and Component 2 needs particular explanation, because the individual theses describe it from two different angles. C3 was designed to supply its category trust score to C2 as the rating input. C2 was designed to accept a rating on a 1 to 5 scale and to apply its own cold-start rule.

In the current release, C2 ranks workers using the **plain star average** of each worker's verified reviews in the requested category, on the 1 to 5 scale. This average is normalised and multiplied by C2's derived rating weight of 0.305. A worker with fewer than three reviews receives the neutral value of 3.5. C3 runs alongside this: every verified review is analysed, and the worker's category trust score (0 to 10), together with the review count and credibility factor, is recomputed and stored in MongoDB Atlas, where it is available through C3's API.

Using the star average for ranking in the first release has two practical advantages. It keeps the ranking input identical to the input on which C2's survey respondents based their judgements, since the survey profiles showed a five-point customer rating. And it isolates the ranking from the known limitations of C3's sentiment model, which was trained on augmented tweets and has not yet been validated on real service reviews (Section 6.6).

Feeding C3's trust score into C2 is the main planned integration step. It requires two reconciliations, which the group has identified:

- **Scale.** C3's 0 to 10 score must be mapped to C2's 1 to 5 normalisation range, for example by halving it. A trust score of 6.0 would then correspond to a rating of 3.0.
- **Cold-start handling.** Both components currently protect new workers, in different ways. C3 shrinks scores towards a prior of 6.0 using $C = N/(N + 5)$,

and C2 substitutes 3.5 below three reviews. Applying both would protect new workers twice. When C3's score is used, C2's substitution should be switched off for workers who have a C3 score, and the group must agree whether the neutral point is 3.0 or 3.5 on the five-point scale.

4.2.4 Shared data model

The backend's MongoDB Atlas database holds the shared data model. Table 4.3 lists the main entities and which components read or write them.

Table 4.3: Main entities of the shared data model

Entity	Main fields	Used by
Worker	Worker identifier, name, worker category (39 labels), worker type, NVQ level, years of experience, availability status, current GPS location	Backend (write); C2 (read, via request); C4 transport (location, via application)
Job / booking	Job identifier, customer, worker, worker category, service and specification, status (including completed), price range, transport fee	Backend (write); C3 (category of review)
Review	Job, worker, customer, category, star rating, optional text, predicted label, confidence, sentiment value	Backend and C3 (write); C2 (star average)
Category score	Worker, category, review count, average stars, mean sentiment, raw score, credibility factor, final trust score	C3 (write); available to the backend and C2

Two privacy rules are built into this model. C1 receives only the problem text, never the customer's identity or location. C3 shares only aggregated scores with other components, never reviewer identities or review text.

4.2.5 Validation rules and error-handling responsibilities

Each service validates its own input and returns a clear error rather than failing silently. C2 rejects an empty worker list with status 400 and out-of-range field values with status 422 and a field-level message, and returns an empty result, not an error, when no worker matches the category. C4's transport engine rejects out-of-bounds coordinates and missing keys, and its estimator encodes unseen category values safely rather than failing. C1 rejects empty or missing text. The backend is responsible for

checking that a job is completed before a review is accepted, and for deciding what the user sees when a service returns an error.

Versioning. Each service exposes its active model or parameters as versioned artefacts: C1's classifier and label encoder files, C2's serialised weight bundle (published through GET /weights), C3's model folder, and C4's model pipeline and environment-variable tariffs. A model can therefore be retrained and redeployed without changing its interface. Formal API version numbers in the URL are not yet used and are listed as future work.

4.3 Shared nonfunctional requirements

The individual theses framed their non-functional requirements using ISO/IEC 25010 [49]. Table 4.4 consolidates them into requirements for the complete solution and shows how each is met.

Table 4.4: Shared non-functional requirements

Quality characteristic	Requirement for the complete solution	How it is addressed
Performance efficiency	Each intelligent step should respond fast enough for an interactive mobile application; ranking within 2 seconds	C2 ranks 10,000 candidates in 4.62 ms (95th percentile); C4 estimator mean latency 11.10 ms; all models run on CPU. End-to-end latency on mobile networks is not yet measured
Usability	Customers must not need trade names, must be able to write in English or Singlish, and must see prices before committing	Free-text search (C1); itemised fee and labelled estimate (C4); readable 0–10 and 0–100 scales
Reliability	A failure in one service or external API must not crash the application	Independent services; input validation; C4 weather timeout falls back to a neutral multiplier; C1 declines low-confidence queries
Security	API keys must stay on the server; services should be reachable only by authorised callers	C4 keys held in server environment variables; C1 and C3 receive no unnecessary personal data. Gateway authentication and rate limiting are pending
Privacy	Personal data must be minimised and processed only for its purpose, in line with the Personal Data Protection Act [50]	C1 sees only problem text; C2 retains nothing after responding; C3 shares only aggregated scores; C4 does not store locations

Quality characteristic	Requirement for the complete solution	How it is addressed
Maintainability	Models, weights and tariffs must be replaceable without code changes	Versioned model files; C2 weight bundle; C3 scoring constants; C4 environment variables for fuel price and tariffs
Scalability	Services must scale independently with demand	All services are stateless and can run as several instances behind a load balancer
Explainability (project requirement)	Every score or price shown to users must be decomposable	C1 confidence; C2 per-criterion breakdown; C3 stored intermediate values; C4 itemised breakdown

4.4 Integrated workflow and exception handling

4.4.1 Normal operation

Figure 4.1 shows the sequence of calls during one complete customer journey, from the search to the review.

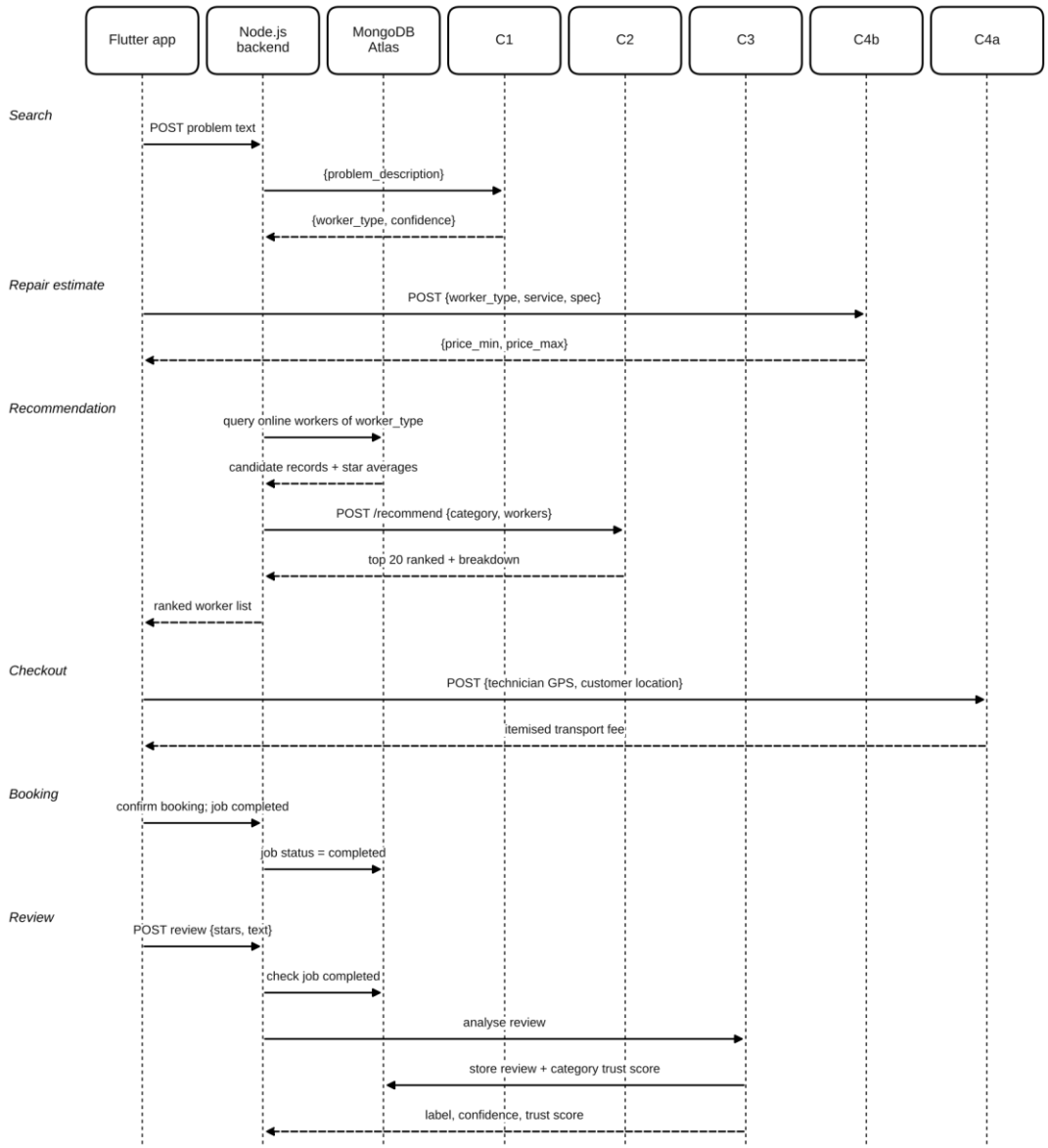


Figure 4.1: Sequence of interactions between the application, backend, database and services

In normal operation, only the backend talks to C1, C2 and C3, and only the application talks to C4. The backend is the only component that writes jobs and reviews to the database, apart from C3's category score records. Each service call is independent, so a slow response from one service delays only the step that needs it.

4.4.2 Failure paths and fallbacks

Table 4.5 lists the important failure paths and the behaviour of the integrated system in each case.

Table 4.5: Exception handling in the integrated workflow

Failure	Detection	System behaviour and recovery
Problem text is ambiguous or unrelated to household services	C1 top probability below 0.4	No category is returned; the application shows "We couldn't find the worker, try again" and the customer rephrases
Problem text is empty or malformed	C1 input validation	Validation error; the service keeps running
No online worker in the predicted category	C2 filter leaves no eligible worker	C2 returns status 200 with an empty list; the application tells the customer that no worker is currently available
Worker record has an out-of-range value	C2 schema validation	Status 422 identifying the field; within scoring, out-of-range values are clamped so a score always stays between 0 and 100
New worker with fewer than three reviews	C2 review count check	Neutral rating of 3.5 is used, so the worker stays discoverable
Weather API does not respond	C4 timeout	Neutral weather multiplier of 1.00 is applied; the fee is still returned
Unseen service or specification sent to the estimator	C4 one-hot encoder (handle_unknown)	Unknown values are encoded as zeros; a valid range with minimum not above maximum is still returned
Invalid coordinates or missing origin or destination	C4 input validation	Error response; no fee is calculated
Review submitted for a job that is not completed	Backend job status check	Review rejected before it reaches C3
Review has stars but no text	C3	Score computed from stars alone
Service idle on a free hosting tier	First request is slow	Application sends a health request at start-up to wake the services early

Two failure paths are not yet handled automatically and are listed as future work. If the Google Maps Distance Matrix API fails, C4 cannot currently estimate a distance; a planned fallback uses the straight-line distance multiplied by a road circuity factor and marks the fee as estimated. And no service currently retries a failed call to another service; the application relies on the user repeating the action.

4.5 Team contribution and dependency matrix

Table 4.6 maps each team member to their component, research tasks, development tasks, integration work and evidence. Table 4.7 shows the dependencies between components. The agreed declaration of contributions is given in Appendix B.

Table 4.6: Team contribution matrix

Member	Research tasks	Development tasks	Integration work	Evidence
Bandara W A M S IT22551320 C1	Code-mixed NLP and intent classification review; 39-category taxonomy; 21,294-row dataset; FAISS vs SVM comparison; error analysis	Training notebooks; FastAPI prediction service with 0.4 rejection rule	Defined the worker_type contract and shared label list; connected search to the backend and C2	C1 thesis; notebooks; model artefacts; test cases T1–T12
Shiwanthika M M IT22283658 C2	MCDM and weight elicitation review; survey instrument; 468-observation regression; stated vs revealed preference analysis	Four-stage ranking engine; FastAPI service with explain endpoint; evaluation console	Category filter on C1 labels; rating input from verified reviews; candidate-in-request design	C2 thesis; survey dataset; regression output; test suite; weight bundle
Wijebandara W.N.S.K IT22643704 C3	Sinhala and code-mixed sentiment review; augmented 101,217-row dataset; XLM-R fine-tuning; trust-score model	FastAPI service with model and scoring logic; MongoDB score records	Verified-completion rule with the backend; category-specific scores; interface to C2	C3 thesis; training notebook; scoring scenarios; integration test
Kumara A.J.T.M IT22137982 C4	Pricing, credence goods and travel-time review; repair price survey and 630-row dataset; DT vs RF comparison	Flask transport fee engine; FastAPI repair estimator; Docker; 14 automated tests	Repair dataset on C1 categories; transport origin from C2's selected worker; checkout integration	C4 thesis; dataset; test suite; deployed endpoints
All members (shared)	Common problem definition; stakeholder analysis	Flutter application; Node.js backend; MongoDB Atlas schema	End-to-end testing; cloud deployment	This report; application; repository history

Table 4.7: Dependency matrix (row depends on column)

Depends on	C1	C2	C3	C4
C1	–	None	None	None
C2	worker_type as category filter	–	Verified review data (star average now; trust score planned)	None
C3	Category of the job, which originates from C1	None	–	None
C4	worker_type for the repair estimate	Selected worker's location for the transport fee	None (trust is deliberately not a pricing input)	–

C1 has no dependencies and is depended on by all others, which is why its label list was fixed first. C4 deliberately does not use trust scores: a highly rated technician is not automatically priced higher, which keeps the transport fee tied to travel conditions and the repair range tied to the job.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

Table 5.1 lists the shared technology stack and the reason for each major choice.

Table 5.1: Common technology stack

Layer	Technology	Reason for the choice
Mobile client	Flutter [33]	One code base for customer and worker interfaces on Android and iOS
Backend	Node.js	Business logic, orchestration of the AI services, and a future API gateway; widely used for JSON APIs
Database	MongoDB Atlas [34]	Managed document database suited to flexible worker, job and review records; scales without application changes
AI services	Python with FastAPI (C1, C2, C3, C4 estimator) and Flask (C4 transport engine)	The machine-learning stacks are in Python; FastAPI gives automatic schema validation and OpenAPI documentation; Flask suits the simple synchronous fee calculation
Machine learning	sentence-transformers, scikit-learn, FAISS (C1); pandas, statsmodels, scipy (C2); PyTorch and Hugging Face Transformers (C3); scikit-learn (C4)	Mature open-source libraries under permissive licences that allow commercial use
Model training	Google Colab notebooks (CPU and free T4 GPU)	No local GPU was available; C3's XLM-R model trains in about 80 minutes on a T4 GPU
External APIs	Google Maps Distance Matrix; OpenWeatherMap	Live road distance, traffic duration and weather for the transport fee
Hosting	Independent cloud web services for each component (C4 on Render, with a Docker container for the estimator)	Low-cost platform-as-a-service hosting with automatic redeployment and environment-based configuration
Version control	Git	Source history for each component and for the shared application

The most important shared infrastructure decision was to keep the AI services stateless and independent of the database, apart from C3's score records. Services receive everything they need in the request and keep only their model in memory. This made each service easy to test alone, easy to redeploy, and easy to scale.

5.2 Development and integration process

The project was carried out in three phases, which overlapped in time.

- **Phase 1, research.** Each member reviewed the literature for their subproblem, defined their research questions, and collected or generated data: C1's household problem descriptions, C2's practitioner survey, C3's augmented sentiment dataset and C4's repair price survey.
- **Phase 2, development.** Each member built and evaluated their model or algorithm and wrapped it as a service. In parallel, the group built the shared Flutter application, Node.js backend and database schema.
- **Phase 3, integration, evaluation and writing.** The services were connected to the application, deployed to the cloud and tested end to end, and the individual theses and this report were written.

Integration followed the interface-first approach described in Section 4.1. The group agreed the category labels and the request and response formats before the services were complete, so that each member could build against the agreed contract. Because each service could be called with synthetic data, components did not have to wait for each other: C2's ranking engine, for example, was validated with up to 50,000 synthetic workers before real worker records existed.

Weight and model updates were designed so that they do not require code changes. C2's weights were re-derived and redeployed when the survey sample grew from an initial partial collection to the final 39 valid respondents, by regenerating its weight bundle. C4's petrol price is updated in the hosting platform's environment settings when CEYPETCO announces a revision. C1 and C3 can be retrained in their notebooks and redeployed by replacing their model files.

5.3 Deployment and operational configuration

All four intelligent services now run in the cloud and are connected to the application through their APIs when needed. Table 5.2 summarises the deployment and operational configuration.

Table 5.2: Deployment and operational configuration

Service	Runtime	Configuration	Operational notes
C1 Intent Recognition	Python FastAPI service; loads embedding model, SVM and label encoder at start-up	Model artefacts as versioned files; 0.4 threshold	CPU-only; stateless; logs of predictions and rejections recommended for retraining
C2 Worker Recommendation	Python FastAPI on Uvicorn; health check path configured	Weights, directions, bounds and cold-start constants in a serialised bundle, published through GET /weights	Free tier may add up to about 90 s on the first request after inactivity; health request at app start mitigates this
C3 Review Credibility	Python FastAPI; loads fine-tuned XLM-R and tokenizer once; uses GPU if available, otherwise CPU	Model folder; scoring constants (60/40, prior 6.0, prior weight 5)	Writes category scores to MongoDB Atlas; low-confidence predictions should be logged for review
C4 Transport Fee	Flask on Gunicorn, Render	BASE_FARE, PER_KM_RATE, CURRENT_PETROL_PRICE and API keys in environment variables	One Distance Matrix call and one weather call per estimate; called only after a worker is selected to limit API cost
C4 Repair Estimator	FastAPI on Uvicorn in a Docker container, Render	Serialised scikit-learn pipeline	Mean inference latency 11.10 ms on a running service
Backend and database	Node.js; MongoDB Atlas	Service URLs and secrets held server-side	Managed database with built-in replication

Configuration. Secrets and tariff values are held in environment variables on the server, never in the mobile application, following the Twelve-Factor principle of keeping configuration in the environment [51]. This allows API keys to be rotated and fuel prices to be updated without releasing a new version of the application.

Monitoring. Each hosting platform's service logs are used to check errors. The individual theses recommend structured logging of four events for operation at scale: C1 predictions and rejections, C3 low-confidence predictions, C4 external API failures and fallback use, and request latency for every service. These logs, kept without user identity, would also provide the first real data for retraining.

Backup and restore. Persistent data lives only in MongoDB Atlas, which provides managed backups. The AI services hold no persistent state, so a failed service is restored simply by redeploying it from its repository and model files. Because C3 stores intermediate values (star average, mean sentiment, review count) as well as the

final score, every trust score can be recomputed if the scoring constants or the model change.

5.4 Evidence of team execution

This section summarises the evidence that the project was planned and executed as a team. The evidence itself is held in MS Planner, MS Teams, the Git repositories and the research diaries. Figures 5.1 and 5.2 show the MS Planner board and progress chart that the team used to plan and track project tasks, and Figure 5.3 shows the repository structure of the integrated solution.

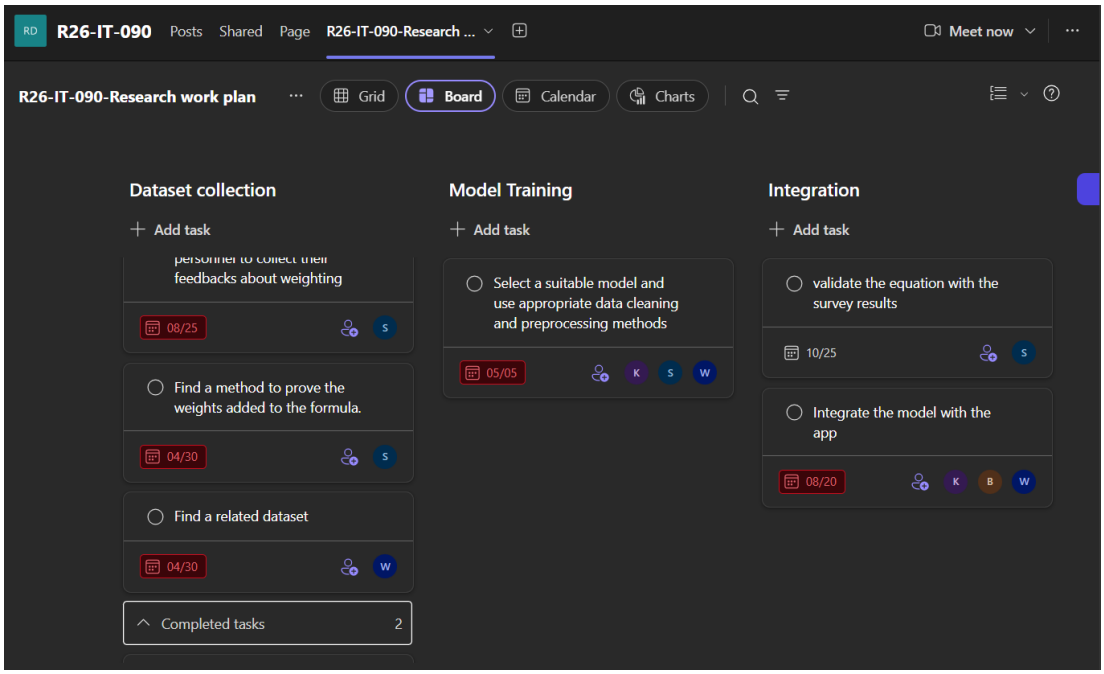


Figure 5.1: MS Planner board showing project tasks, owners and status

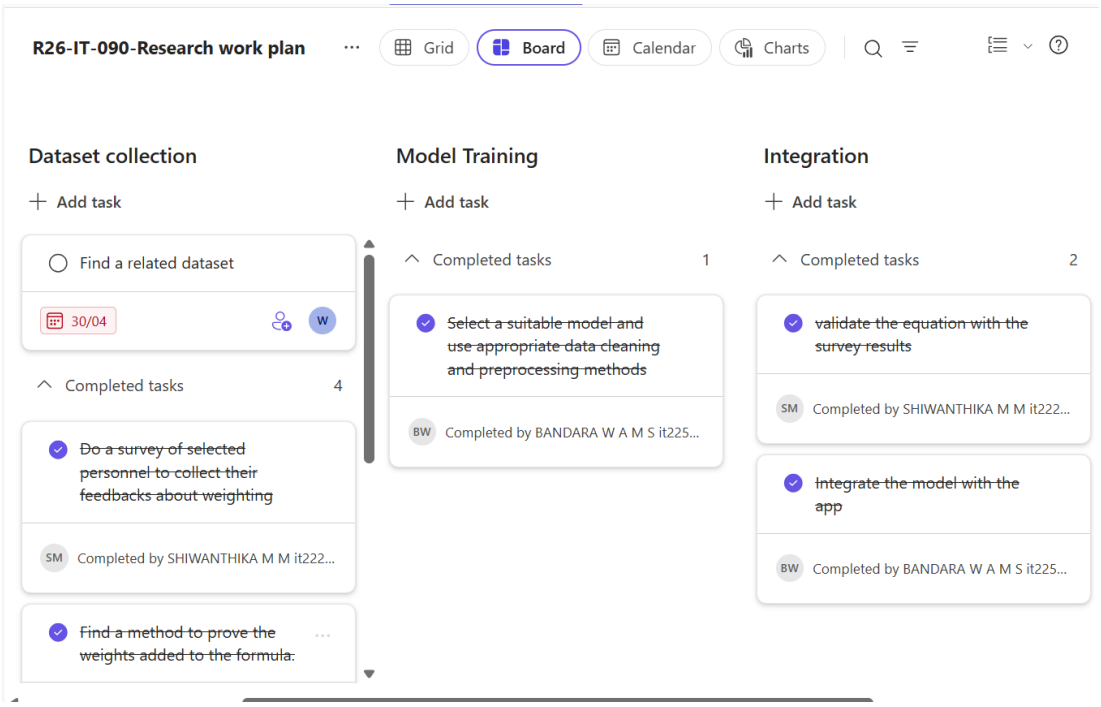


Figure 5.2: MS Planner progress chart for the project

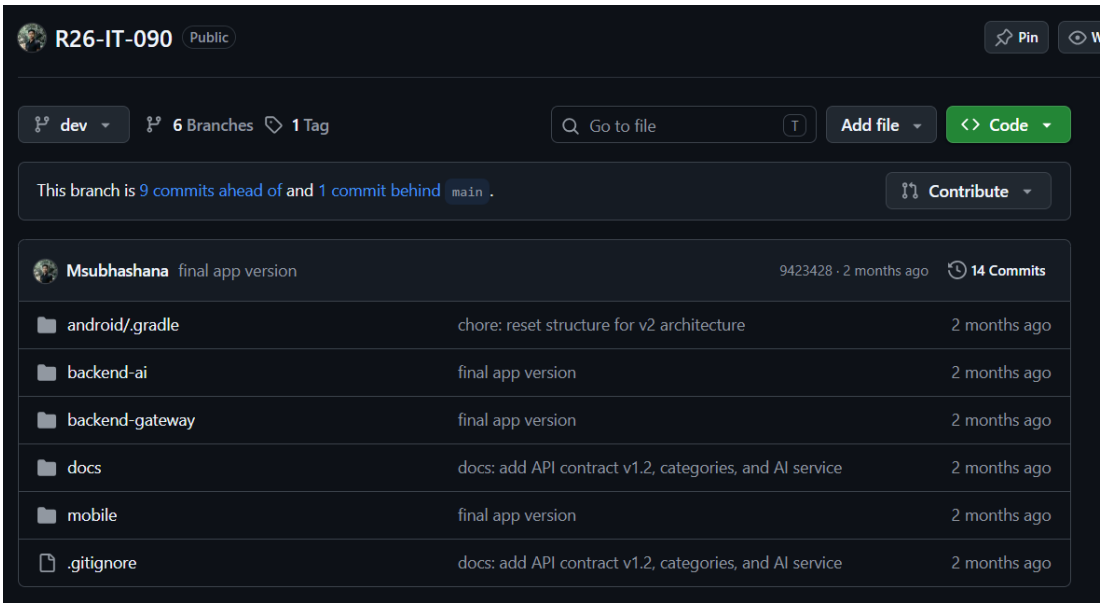


Figure 5.3: Repository structure of the integrated solution

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Evaluation was organised at two levels. At the **component level**, each member evaluated their own model or algorithm against a baseline or a target, as reported in the individual theses. At the **system level**, the group evaluated whether the components work together as one product and whether the group objectives of Section 2.4 were met. Testing followed the separation of test levels in ISO/IEC/IEEE 29119 [52]: unit and functional tests of each service, interface tests between services, and system tests of the complete journey.

The group-level evaluation questions were:

- **EQ1.** Does each intelligent decision meet its success criterion against a baseline or target?
- **EQ2.** Do the components exchange data correctly through their agreed contracts?
- **EQ3.** Does the complete customer journey run end to end in the cloud-deployed system?
- **EQ4.** Do stakeholders see value in the solution, and what concerns remain?

Table 6.1 defines the acceptance criteria, baselines, metrics and data used for each group objective, set before the results are presented.

Table 6.1: Evaluation framework and success criteria

ID	Acceptance criterion	Baseline or target	Metric and data
GO1	Accuracy and macro-F1 clearly above baseline; low-confidence queries rejected	FAISS top-1 nearest neighbour on the same embeddings and split	Accuracy with 95% CI, macro-F1, per-category F1; 4,259-row held-out set
GO2	Significant, empirically derived weights; bounded, monotonic, deterministic and decomposable scores; ranked list within 2 s; new workers discoverable	Direct importance ranking by the same respondents; NFR target of 2 s	Regression coefficients and p-values (468 observations); property tests; latency for pools of 100 to 50,000; cold-start score difference

ID	Acceptance criterion	Baseline or target	Metric and data
GO3	Sentiment accuracy and weighted F1 of at least 85%; scores that behave as designed for few, many and conflicting reviews	85% target	Accuracy, weighted and per-class F1 on 16,853 held-out rows; eight scoring scenarios
GO4	Price range accuracy comparable to an ensemble baseline; fee transparent, monotonic and bounded; services robust and fast	100-tree random forest; 15 ms mean latency target	MAE, RMSE and R^2 on hold-out split; scenario analysis; 14 automated tests; inference latency
GO5	All interfaces exchange valid data; complete journey succeeds in the deployed system	–	Interface tests; end-to-end test cases E1–E10
GO6	Stakeholder evidence collected; commercial, ethical and SDG analysis completed	–	Practitioner survey (n = 41); Chapter 7

6.2 Integration and end to end testing

6.2.1 Interface tests

Interface tests checked that each pair of connected components exchanges valid data. Table 6.2 summarises them.

Table 6.2: Interface test results

ID	Interface	Test and expected result	Result
I1	C1 → C2	A worker_type returned by C1 is applied unchanged as C2's category filter; only workers of that category are returned from a mixed-category pool	Pass
I2	C2 filter	A pool of nine workers including one offline worker and one of another category returns seven eligible and seven ranked workers	Pass
I3	Verified reviews → C2	Star ratings in the 1 to 5 range are normalised correctly; workers with fewer than three reviews receive the neutral rating	Pass
I4	Backend → C3	A review for a completed job is classified, stored with its sentiment value, and the worker's category score is updated	Pass

ID	Interface	Test and expected result	Result
I5	C1 → C4 estimator	A worker category from C1 is accepted by the repair estimator; unseen values are handled without error and price_min ≤ price_max	Pass
I6	C2 → C4 transport	The selected worker's latitude and longitude are sent as the transport origin and an itemised fee is returned	Pass
I7	C4 robustness	Invalid coordinates and missing keys return errors; a simulated weather timeout falls back to the neutral multiplier (part of 14 automated tests)	Pass (14/14)

6.2.2 End-to-end test of the customer journey

After all four services were deployed to the cloud, the team ran the complete customer journey in the mobile application. The journey completed successfully. Table 6.3 lists the end-to-end test cases, which follow the steps of Figure 3.2.

Table 6.3: End-to-end test cases for the complete customer journey

ID	Step and input	Expected result	Result
E1	Customer enters a Singlish problem, e.g. "watura bata pupurala"	C1 returns a plumbing-related worker category with confidence of at least 0.4	Pass
E2	Customer enters an unrelated or ambiguous query	Application shows "We couldn't find the worker, try again"	Pass
E3	Customer selects a service and specification for the identified trade	A repair price range in LKR is displayed and labelled as an estimate	Pass
E4	Worker list requested for the identified category	Up to 20 online workers of that category are listed in descending score order	Pass
E5	Customer selects a worker and proceeds to checkout	An itemised transport fee based on the worker's location and current conditions is displayed	Pass
E6	Customer confirms the booking; job is marked completed	Job status stored as completed in MongoDB Atlas	Pass
E7	Customer opens the review screen for a job that is not completed	Review is not accepted	Pass
E8	Customer submits stars and Singlish text for the completed job	C3 classifies the text and stores the review and the updated category trust score	Pass

ID	Step and input	Expected result	Result
E9	Customer submits stars without text	Score is updated from stars alone	Pass
E10	A new search in the same category after the review	The worker's updated star average is used in C2's ranking	Pass

The following figures provide visual evidence of the end-to-end run. Figures 6.1 to 6.7 show the application at each step of the journey. In the application, the category trust score from C3 is displayed on a 0 to 100 scale: for example, “Trust 63” in Figure 6.7 corresponds to a trust score of 6.3 out of 10, and a worker with no reviews in a category is labelled “New”.

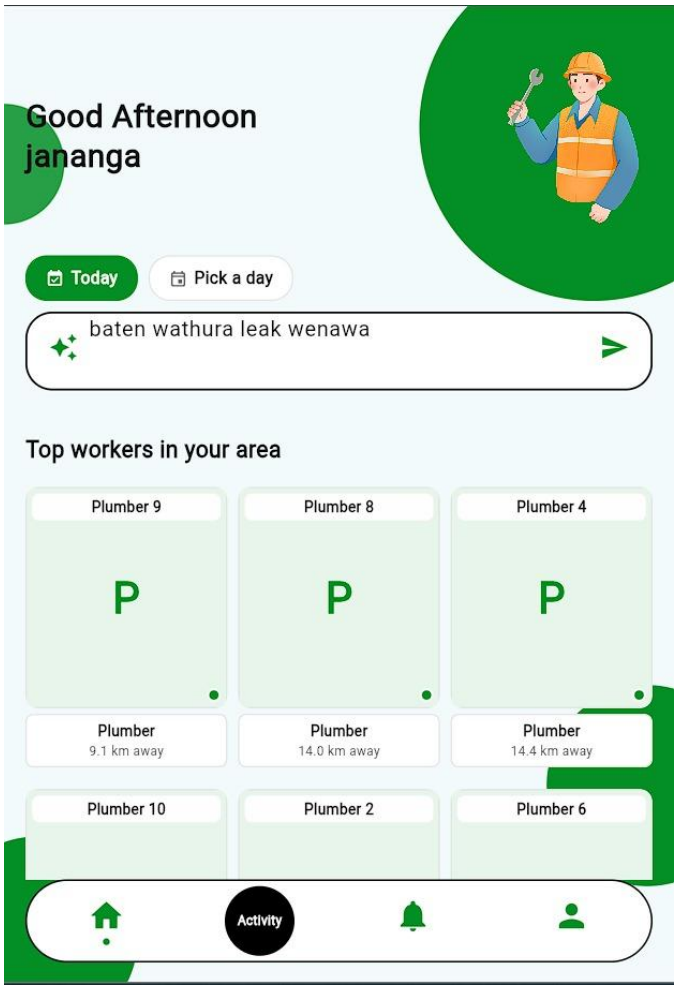


Figure 6.1: Free-text search with a Singlish problem description

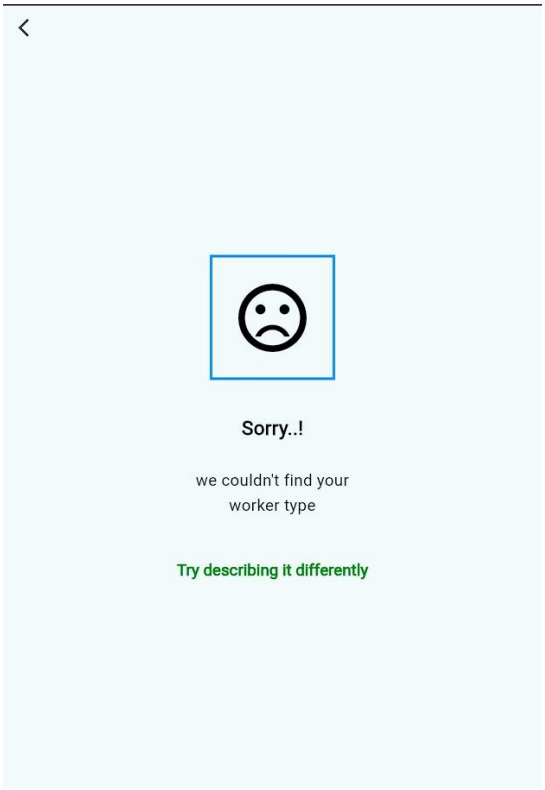


Figure 6.2: Low-confidence message for an unrelated query

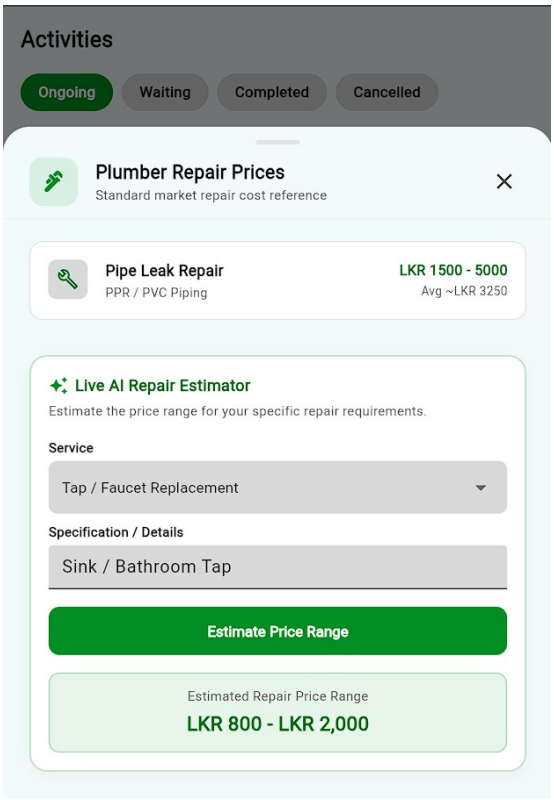


Figure 6.3: Repair price range displayed for a selected service

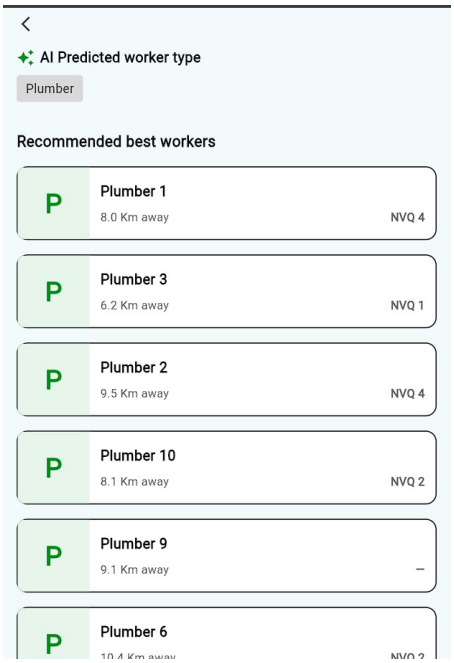


Figure 6.4: Ranked list of recommended workers

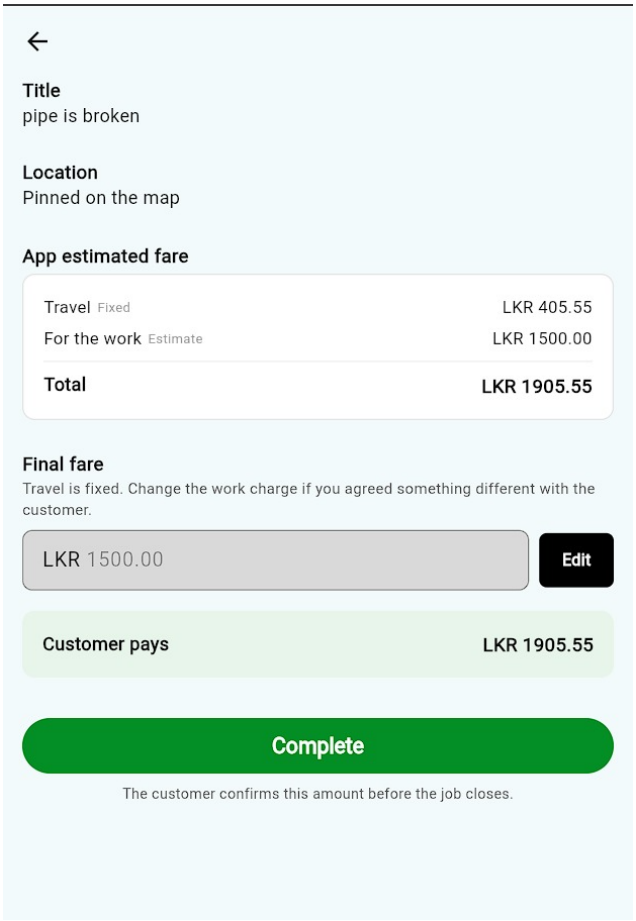


Figure 6.5: Checkout with itemised transport fee

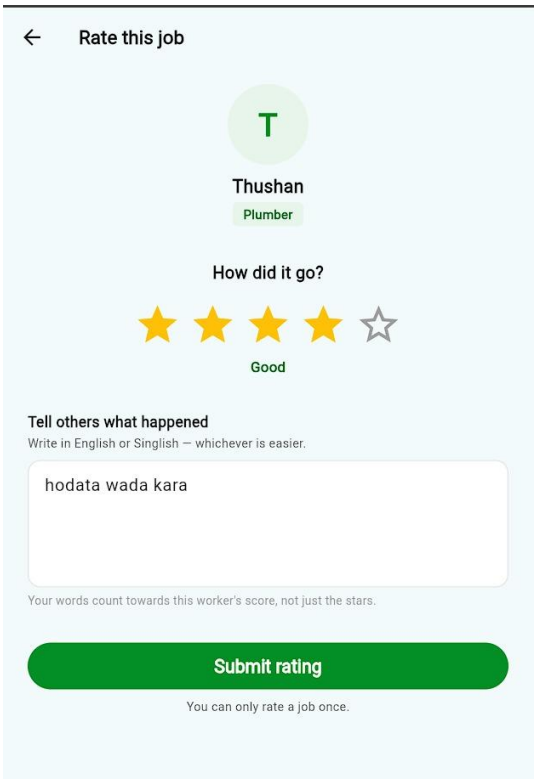


Figure 6.6: Review screen for a completed job

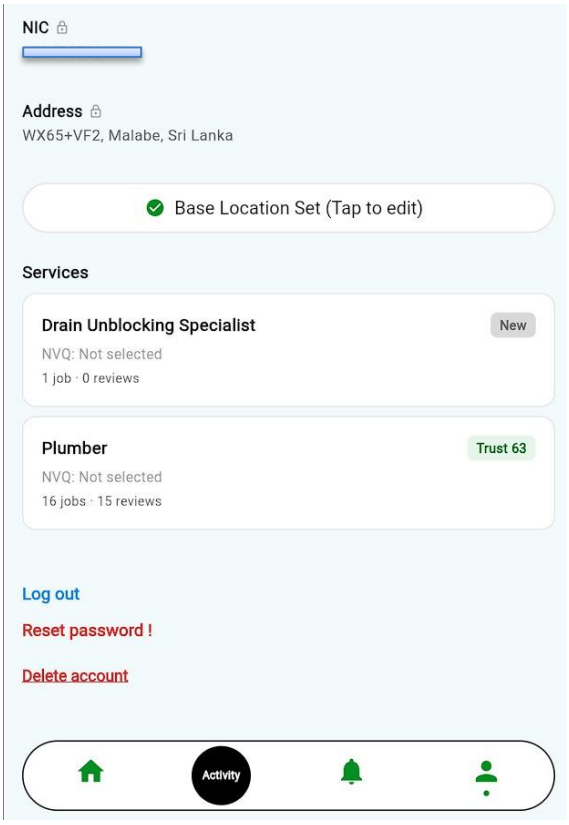


Figure 6.7: Updated trust score after a review

6.2.3 Non-functional and other tests

Performance was measured at component level. C2's ranking latency was measured for candidate pools from 100 to 50,000 workers, and C4's estimator latency was measured on the deployed service. Security testing at system level was limited to checking that API keys are held on the server; authentication and rate limiting are pending. Regression testing was carried out manually when services were redeployed. The following were not carried out and are reported as limitations: load testing of the complete system under many simultaneous users, measurement of end-to-end response time on mobile networks, and formal usability testing with real users.

6.3 Overall results

6.3.1 Component results in the context of the whole

Table 6.4 brings together the principal results of the four components. Detailed experiments, per-category results and error analyses are reported in the individual theses.

Table 6.4: Principal results of the four components

Component	Data	Principal result	Main caveat
C1 Intent Recognition	21,294 descriptions; 39 categories; ~42% Singlish	SVM 78.68% accuracy (95% CI 77.45–79.91%), macro-F1 0.79; FAISS 71.50%, 0.71; F1 improved in 36 of 39 categories; 78.35% on cleaned data	Mostly AI-generated data; near-duplicates may make the test easier than real use
C2 Worker Recommendation	41 respondents, 39 valid, 468 evaluations	Weights: proximity 0.396, rating 0.305, experience 0.215, NVQ 0.084; R^2 0.200; three of four coefficients $p < 0.001$; stated vs revealed $\rho = -0.400$; 4.62 ms for 10,000 workers (95th percentile)	NVQ coefficient not significant ($p = 0.092$); sample weighted towards the Western Province and less experienced professionals
C3 Review Credibility	101,217 rows; 67,411 train, 16,853 test	95.41% accuracy; weighted F1 0.954; per-class F1 0.951–0.958; scoring scenarios behave as designed	Augmented tweets, not service reviews; possible leakage; overconfidence on neutral text

Component	Data	Principal result	Main caveat
C4 Pricing Engine	105 surveyed combinations, 630 rows; live API inputs	Decision tree R^2 0.888 / 0.925, MAE LKR 1,712 / 3,144, comparable to random forest; fee monotonic, bounded at ×1.755, +7.4% across 2026 fuel prices for 10 km; 14/14 tests; 11.10 ms mean latency	Augmented copies across the split make accuracy optimistic; no user validation of the fee

6.3.2 Synthesised system-level results

Read together, the results support five system-level findings.

First, every step of the customer journey now has a working, evaluated intelligent decision. Each component met or exceeded its own success criterion: C1 beat its baseline by 7.18 accuracy points with non-overlapping confidence intervals; C2's ranking satisfied all of its property requirements and ran more than four hundred times faster than its 2-second requirement; C3 exceeded its 85% accuracy target; and C4's decision tree matched the random forest baseline while keeping its explanation as rules.

Second, the components work together. All interface tests passed, and the complete journey from a free-text Singlish search to a stored trust score ran successfully in the cloud-deployed system. The shared 39-label vocabulary meant that no mapping layer was needed between C1's prediction, C2's filter, C3's category records and C4's price dataset, apart from the conversion between machine and readable label forms.

Third, the intelligent steps add negligible delay. The measured computation times are milliseconds: about 5 ms for C2 to rank 10,000 candidates and about 11 ms for C4 to estimate a price. C1 and C3 run on CPU without a GPU. The delay that users notice is therefore dominated by network round trips, by the external routing and weather APIs, and by the wake-up time of idle free-tier services, not by the models.

Fourth, explainability is consistent across the system. Every number a user sees can be traced: a confidence for the category, a per-criterion breakdown for the worker score, stored intermediate values for the trust score, and an itemised transport fee. This consistency is a property of the integrated design, not of any single component.

Fifth, the evidence is strongest for technical feasibility and weakest for real-world accuracy. Three of the four components were trained or evaluated on synthetic, augmented or derived data, and the system has not yet been used by real customers and workers. The results show that the solution can be built and works as designed; they do not yet show how accurately it performs on real queries, reviews and prices.

6.4 Stakeholder and user validation

Stakeholder evidence was collected at two points.

Hiring professionals (C2). Forty-one professionals who hire or assign household maintenance workers, including 11 service company owners or managers, 8 contractors or site supervisors and 6 human resource managers, completed the profile-based survey. Their judgements determined the ranking weights used in the deployed system. The survey also produced an important finding for the whole project: when asked directly, the professionals ranked distance as the least important criterion, but their actual evaluations weighted it most heavily. This shows that stakeholder statements alone are not a reliable basis for parameterising the system, and supports the group's preference for behavioural and data-driven evidence.

Tradespeople and workshops (C4). Independent tradespeople and repair workshops supplied typical labour charges and quotations, which form the 105 base combinations of the repair price dataset. They took part as data sources and did not evaluate the system.

Changes made in response to stakeholder evidence. The C2 survey led directly to the replacement of researcher-assigned weights with practitioner-derived weights, and to the retention of the NVQ criterion on policy grounds. The C4 survey grounded the repair price ranges in local market rates instead of assumed prices.

Remaining concerns. The application has not yet been used by real customers and workers. The group has therefore not yet measured whether real customers' descriptions are understood as accurately as the synthetic test data, whether customers find the trust scores and prices fair, or whether technicians accept more jobs with the context-aware transport fee. A field trial with real users is the most important future work (Section 8.3).

6.5 Achievement of common objectives

Table 6.5 maps the evidence to each group objective.

Table 6.5: Achievement of the group objectives

ID	Status	Justification
GO1	Achieved, with limitations	SVM accuracy 78.68% and macro-F1 0.79 against 71.50% and 0.71 for the baseline, with non-overlapping confidence intervals; rejection rule works in the deployed system. Limitation: accuracy measured on largely AI-generated data
GO2	Achieved	Weights derived from 468 practitioner judgements; scores bounded, monotonic, deterministic and decomposable; 10,000 candidates ranked in 4.62 ms; cold-start rule keeps new workers discoverable. The NVQ coefficient was not significant and is retained on stated policy grounds
GO3	Achieved, with limitations	95.41% accuracy against the 85% target; scoring behaves as designed; verified-completion rule enforced. Limitations: optimistic accuracy due to augmented tweet data; trust score not yet used as C2's ranking input
GO4	Achieved, with limitations	Decision tree comparable to random forest (R^2 0.888 and 0.925); transparent, bounded and fuel-indexed transport fee; 14/14 tests; 11.10 ms mean latency. Limitation: leakage across the split makes accuracy optimistic
GO5	Achieved	All four services deployed in the cloud; shared vocabulary and documented contracts; interface tests passed; complete journey ran successfully. Gateway authentication, load testing and end-to-end latency measurement remain open
GO6	Partly achieved	Practitioner survey completed and commercial, ethical and SDG analysis carried out (Chapter 7); validation with real users of the running application has not yet been carried out

6.6 Integrated limitations and threats

Dependence on synthetic and augmented data. C1's descriptions are mostly AI-generated, C3's Sinhala and Singlish text is machine-translated, transliterated or templated from English tweets, and C4's 630 rows were expanded from 105 surveyed combinations. In each case the test data comes from the same process as the training data, and in C3 and C4 augmented versions of the same item may appear on both sides of the split [53]. The reported accuracies are therefore upper bounds for real use. This is the largest validity threat to the system as a whole, because errors compound along the journey: a wrong category from C1 sends C2 to the wrong pool of workers, C4 to the wrong price list, and C3's review to the wrong reputation record.

Error propagation from Component 1. About one in five synthetic test descriptions is assigned to a wrong category, and the confusions concentrate in neighbouring trades. Some of these errors matter little in practice (a furniture restorer can often handle simple carpentry); others, such as a CCTV problem sent to a general electronics technician, would produce irrelevant recommendations and prices. The 0.4 rejection threshold reduces this risk but has not been tuned on real queries.

Partial use of reputation in ranking. The trust score built by C3 is computed and stored but is not yet the rating input to C2. Until it is, the ranking does not benefit from review text or from C3's credibility adjustment, and the two components handle new workers in different ways (Section 4.2.3).

Equity effect of the proximity weight. Proximity carries the largest weight in C2's ranking. This faithfully reflects practitioner judgement but structurally disadvantages workers outside dense service areas. Combined with a transport fee that rises with distance, a distant worker is both ranked lower and priced higher. The group regards this as a system-level fairness risk that needs mitigation (Section 8.3).

Untuned constants. Several constants that shape user-facing results were set by judgement rather than estimated from data: C1's 0.4 threshold, C3's 60/40 fusion weights, prior of 6.0 and prior weight of 5, and C4's base fare, distance rate and multipliers.

Generalisability. Practitioner judgements, price data and the first target market are concentrated in the Western Province, and the language support excludes Tamil and Tanglish. The results should not be assumed to transfer to other regions or language groups without new data.

Unresolved integration risks. C4's endpoints are not yet behind an authenticated gateway, there is no automatic fallback when the routing API fails, free-tier hosting can delay the first request by up to about ninety seconds, and the complete system has not been load tested. These must be resolved before a public launch.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The marketplace creates value for each group in the two-sided market. Its value comes from the combination of the four components, because a customer gains little from a correct category if the recommended workers cannot be trusted or the price is unknown. Table 7.1 summarises the value proposition by beneficiary.

Table 7.1: Value proposition by beneficiary

Beneficiary	Pain point today	Value delivered by the integrated solution
Household customers	Do not know the trade name; cannot judge which worker is suitable or trustworthy; fear over-charging; travel charges negotiated ad hoc	Search in their own words, including Singlish; a ranked, explainable shortlist; trust scores that reflect what other customers wrote; a repair price range before choosing a worker and an itemised transport fee before confirming
Workers and technicians	Depend on personal networks; new workers are invisible; one bad review can be damaging; travel in traffic, rain and periods of high fuel prices is unpaid	Jobs matched to their own trade; visibility from the first day; reputation protected from a single review and separated by trade; a transport fee that rises with distance, congestion, weather and fuel price, passed fully to the worker
Platform operator	Unfilled requests, disputes and opaque rules damage trust	Ranking, trust and pricing rules that are documented, explainable and configurable without code changes
Society	Informal work outside the digital economy; language barriers to digital services	A digital channel for informal tradespeople; support for Sinhala and Singlish; transparent prices for less informed customers

The value proposition is supported by stakeholder evidence in two ways. The practitioner survey confirmed that suitability depends on several criteria whose trade-offs customers would otherwise have to make alone. The tradespeople who supplied prices confirmed that repair prices vary widely and are rarely published.

7.2 Market alternatives and differentiation

Target segments. The first target segment is urban and suburban households in the Western Province, where traffic congestion is most severe, smartphone use is high, and many residents have recently moved and lack trusted contacts. The second segment is the independent technicians and small workshops who serve them.

Secondary segments include landlords and property managers, who handle many different problems, and older users who are less familiar with trade terminology.

Alternatives. Households currently rely on word of mouth, classified-advertisement sites such as ikman.lk, local freelance marketplaces, fixed-menu home service applications and informal social-media groups. International task platforms such as TaskRabbit do not operate in Sri Lanka but set user expectations. Table 2.1 compared these alternatives against the four failures in the customer journey.

Differentiation. The marketplace is differentiated by a combination of features that no alternative offers together:

- free-text understanding of English and romanised Singlish problem descriptions;
- ranking weights derived from practitioner judgement and disclosed, including a national vocational qualification signal;
- trust scores that read review text in English, Sinhala and Singlish, adjust for the number of reviews and are separated by trade;
- a locally grounded repair price range and a transparent, fuel-indexed transport fee instead of demand-based surge pricing.

The differentiation is defensible for three reasons. The datasets (household problem descriptions, practitioner judgements, augmented Singlish sentiment data and Sri Lankan repair prices) took field work and effort that a competitor would need to repeat. The data advantage grows with use, because every completed booking adds a real query, a real review and a real price. And the platform benefits from network effects: fair visibility and fair prices help attract workers, which improves choice and attracts customers [7].

7.3 Business model and feasibility

Table 7.2 summarises the business model of the integrated marketplace using the business model canvas [54].

Table 7.2: Business model canvas of the marketplace

Building block	Content
Customer segments	Urban and suburban households; independent technicians and small workshops; later, property managers and field-service businesses
Value propositions	Find, trust and fairly pay a suitable worker from a description in one's own words (Table 7.1)
Channels	Flutter mobile application; later, business-to-business APIs
Customer relationships	Self-service in the application; explainable scores and itemised prices to build trust; post-job reviews
Revenue streams	Commission on the repair labour charge of each completed job (no commission on the transport fee); optional subscription or featured-listing plans for workers and service companies; later, licensing of pricing and review-analysis APIs and aggregated, anonymised demand insights for partners
Key resources	The four trained models and their datasets; the 39-category taxonomy; the derived ranking weights; the application and backend
Key activities	Collecting real queries, reviews and prices; retraining models; updating fuel prices and tariffs; re-deriving ranking weights periodically; onboarding and verifying workers
Key partners	Cloud hosting providers; Google Maps Platform; OpenWeatherMap; MongoDB Atlas; technician associations and workshops; the Tertiary and Vocational Education Commission for qualification verification
Cost structure	Cloud hosting of four services, backend and database; per-request mapping and weather API charges beyond free allowances; periodic GPU time for retraining; data collection and labelling effort; marketing and worker onboarding

Technical feasibility. The solution is already built, deployed and working end to end. All four services run on CPUs; the heaviest model, C3's XLM-R, can be retrained in about 80 minutes on a free cloud GPU. Ranking and pricing need only milliseconds per request.

Operational feasibility. Running costs are low and predictable. The largest variable cost is external API usage, about one routing call and one weather call per transport estimate. This is kept proportional to real bookings because the fee is requested only after a worker is selected, and it can be reduced further by caching weather by area for a few minutes. Other recurring costs are modest: annual re-administration of the practitioner survey to keep C2's weights current, retraining of C1 and C3 on real data, and periodic refresh of C4's price data.

Sustainability. The model is sustained by a data flywheel. Each completed job can turn the customer's query and final booked category into a new labelled example for C1, the review into real service-review data for C3, and the agreed price into a real transaction for C4. Over time this reduces the system's dependence on synthetic data, which is its main technical weakness.

7.4 Adoption intellectual property and compliance

Adoption barriers. The main barriers are trust in a new platform, worker onboarding, and the "empty marketplace" problem at launch. The design addresses the first two through explainable scores, verified-only reviews and cold-start protection for new workers. The third requires a focused launch in one area and trade group, with active onboarding of workers before customers are invited.

Ownership and licensing. The original assets of the project are the 39-category taxonomy and household problem dataset, the derived ranking weights, the augmented Singlish sentiment dataset and fine-tuned model, the repair price dataset and model, the transport fee model and parameters, and the service code. The software uses open-source components under permissive licences that allow commercial use with attribution, including the Apache 2.0-licensed multilingual embedding model, the MIT-licensed xlm-roberta-base model, and BSD- and MIT-licensed frameworks. The derived ranking weights are published in the Component 2 thesis for reuse. Before commercial release, the group must check the licence of the Kaggle dataset used by C3, the terms of the translation service used for augmentation, the terms of the AI assistant used to generate part of C1's data, and the storage restrictions in the Google Maps and OpenWeatherMap terms of service.

Data protection. Customer queries, worker locations and reviews are personal data under Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [50]. The architecture already applies data minimisation and purpose limitation: C1 sees only the problem text, C2 retains nothing after responding, C3 shares only aggregated scores, and C4 uses locations only to compute the fee and does not store them. Before launch, the platform needs a privacy notice explaining automated processing, a lawful basis for each processing purpose (normally performance of the booking contract), retention

periods, a process for workers to see and question their scores, and safeguards for transfer of location data to overseas API providers.

Consumer protection. Prices shown to consumers should be clear and not misleading, in line with the Consumer Affairs Authority Act, No. 9 of 2003 [55]. The repair range is therefore labelled as an estimate, and the transport fee is shown with its breakdown before the customer confirms.

Responsible AI. The group identified and mitigated the main AI risks in line with the NIST AI Risk Management Framework [56]. Table 7.3 summarises them.

Table 7.3: Responsible AI risks and mitigations

Risk	Where it arises	Mitigation in place or planned
Over-confident wrong predictions	C1 category; C3 sentiment	C1 rejection threshold; C3 credibility shrinkage limits the effect of single reviews; planned calibration [57] and threshold tuning on real data
Unfair exclusion of workers	C2 ranking; C3 reputation	Cold-start rules in both components; decomposable scores that can be explained to a worker; planned reserved shortlist positions for high-quality distant workers
Algorithmic control and opacity	Ranking and pricing	Disclosed weights, itemised fees and no demand-based surge, responding to known criticisms of platform work [58], [23]
Bias in ratings and selection	Reviews and ranking	Verified-completion rule; text-aware scores; monitoring of score distributions across worker groups is planned [59]
Synthetic-data artefacts	C1, C3, C4	Limitations disclosed in every thesis; documentation following datasheet and model-card practice [60], [61]; real-data evaluation planned
Misuse of generative AI in research	Data generation and writing	Disclosure of AI use in each thesis and in Appendix D; all reported figures produced by executed code

Professional responsibility. The group reported negative and partial results openly, including the non-significant NVQ coefficient, the overconfidence of the sentiment

model, and the leakage in the repair price evaluation. A ranking and pricing system affects people's income, so its limitations must be visible to those who rely on it.

7.5 Sustainable Development Goal contribution

Table 7.4 evaluates the plausible contribution of the integrated solution to the selected SDG targets, the possible negative effects, and the measures needed to sustain the benefits.

Table 7.4: Evaluation of the SDG contribution

SDG	Plausible contribution	Possible negative effects	Measures needed
SDG 8 Decent work (target 8.3)	Digital access to customers for informal tradespeople; portable, verified reputation; travel costs compensated and protected against fuel price rises; newcomers kept visible	Proximity weighting may concentrate work among workers in dense areas; platform dependence and algorithmic control	Reserve shortlist positions for high-quality distant workers; disclose ranking rules; let workers query their scores
SDG 9 Innovation and infrastructure	Low-cost, locally adapted AI services that small enterprises could not build themselves	Reliance on foreign cloud and API providers	Keep services portable and stateless; monitor costs and terms
SDG 10 Reduced inequalities	Singlish and Sinhala support lowers the language barrier; same fee for the same trip; prices reduce exploitation of less informed customers	Tamil-speaking users currently excluded; users without smartphones excluded	Extend to Tamil script and Tanglish; consider assisted channels
SDG 11 Sustainable cities	Faster repair of leaks, drains and electrical faults; weather premium keeps service available during storms	Weather premium raises costs for customers in bad weather	Keep the premium bounded and itemised
SDG 12 Responsible consumption	Price ranges can favour repair over replacement; better matching reduces failed repairs and repeat visits	None identified beyond those above	Measure repeat-repair rates once real data exist

These contributions are plausible but not yet measured. They depend on adoption at scale, and on the group's ability to mitigate the equity risk of proximity weighting and to extend language support to Tamil. Measuring them requires the field trial and real-use data described in Section 8.3.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This project set out to address a common problem in Sri Lanka's informal household maintenance sector: customers cannot reliably find, evaluate, trust and fairly pay a suitable worker for a problem they describe in their own words, and capable workers, especially new ones, cannot become visible on fair terms. The evidence in this report shows that an integrated, AI-powered marketplace can address each of the four failures in that journey, and that the four solutions can operate together as one working system.

A customer can now type a problem in English or romanised Singlish and be directed to the correct trade for nearly four in five synthetic test descriptions, with the system declining rather than guessing when unsure. The customer receives a ranked shortlist whose ordering reflects how practitioners actually judge suitability, rather than how they say they do or how a designer assumed. Reputation is built only from verified jobs and can read review text in three language forms without overreacting to a single review. Before anyone comes to the door, the customer sees an indicative repair price and an itemised travel fee that compensates the worker for distance, traffic, weather and fuel prices. All four services run in the cloud, and the team completed the full journey in the application, from search to post-job trust update.

The answer to the common problem is therefore positive on technical feasibility and integration, and provisional on real-world effectiveness. The solution works as designed and is explainable at every step. Its accuracy on real customer queries, real reviews and real prices, and its effect on customer trust and worker income, have not yet been measured, because three of the four components were trained or evaluated on synthetic or augmented data and the application has not yet been used by the public.

8.2 Principal integrated contribution

Beyond the four individual contributions, the team produced four things of value that exist only because the components were designed and built together.

- **An end-to-end, working reference system** for an AI-assisted household services marketplace in Sri Lanka, in which every step of the customer journey

is supported by an evaluated intelligent decision, deployed as independent cloud services behind documented contracts.

- **A shared 39-category vocabulary of household trades** that links language understanding, ranking, reputation and pricing. This taxonomy, and the lesson that categories customers cannot distinguish in words will also be confused by a model, is reusable by any platform in this market.
- **A consistent explainability and fairness design** across the system: a confidence or decomposition behind every number a user sees, and explicit protection for new workers in both ranking and reputation.
- **An integration-level understanding of the system's risks** that no single component reveals: errors from intent recognition propagate to ranking, pricing and reputation; proximity weighting and distance-based fees together disadvantage distant workers; and two components protect new workers in different ways and must be reconciled when the trust score becomes the ranking input.

The team also learned that empirical evidence must come from behaviour and data, not only from statements. C2's finding that professionals' stated priorities contradict their actual judgements is a lesson for the whole project: future tuning of thresholds, weights and constants should be based on observed bookings and outcomes.

8.3 Future directions

The following work is listed in order of priority.

- **Field trial with real users.** Release the application to a limited group of customers and workers in one area of the Western Province. Measure booking completion, customer satisfaction and perceived fairness of scores and prices, and technician acceptance of jobs under the context-aware transport fee.
- **Real-data evaluation of every model.** Build small, hand-labelled test sets of real Singlish and English problem descriptions (C1) and real service reviews in English, Sinhala and Singlish (C3), and compare predicted price ranges with final agreed prices (C4). Re-evaluate C3 and C4 with leakage-free, grouped splits.

- **Complete the reputation integration.** Feed C3's trust score into C2 as the rating input, rescaled to the five-point range, with a single agreed cold-start rule and a single neutral point.
- **Mitigate the proximity bias.** Reserve a defined portion of the shortlist for workers who rank highly on non-proximity criteria but are outside the immediate area, and monitor how work is distributed across areas.
- **Tune constants on real outcomes.** Calibrate C1's rejection threshold against accuracy and coverage, calibrate C3's confidence values, estimate C3's fusion weights and prior from real data, validate C2's weights against actual customer selections, and calibrate C4's tariff parameters with acceptance data.
- **Harden for production.** Route all service calls through the Node.js gateway with authentication and rate limiting; add a routing fallback for C4; move from free hosting tiers to always-on instances; add automated API tests, load tests and end-to-end latency measurement; and introduce API versioning and structured monitoring.
- **Extend coverage.** Add Tamil script and Tanglish support, Singlish spelling normalisation, top-two category suggestions when C1 is uncertain, aspect-based review analysis, region and season features for prices, and personalised ranking learned from customers' selections.

Together, these steps would turn a technically proven, integrated prototype into a marketplace whose benefits for Sri Lankan households and workers can be demonstrated with real evidence.

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APPENDICES

Appendix A Integrated interface specification

This appendix gives the request and response formats of the integrated interfaces summarised in Table 4.2. All messages are UTF-8 JSON over HTTPS. Examples are taken from the individual theses; values are illustrative.

A.1 Component 1: intent recognition

```
Request (from Node.js backend):
{ "problem_description": "watura bata pupurala" }

Response - category found:
{ "worker_type": "plumber", "confidence": 0.894 }

Response - below the 0.4 threshold:
{ "worker_type": null, "confidence": 0.174,
  "message": "We couldn't find the worker, try again." }
```

The value of worker_type is one of the 39 machine labels in Appendix C.

A.2 Component 2: worker recommendation

```
POST /recommend?explain=true
{
  "category": "plumber",
  "top_n": 20,
  "workers": [
    { "id": "W04", "name": "...", "category": "plumber",
      "availability": "Online", "nvq": 7, "rating": 4.7,
      "review_count": 12, "experience": 10, "distance_km": 1.75 }
  ]
}

200 OK
{
  "category": "plumber",
  "total_received": 9, "eligible_after_filter": 7, "returned": 7,
  "results": [
    { "rank": 1, "id": "W04", "final_score": 83.42,
      "cold_start_applied": false,
      "breakdown": {
        "normalized": { "nvq": 100.0, "rating": 92.5,
                        "experience": 40.0, "distance": 96.5 },
        "weighted":   { "nvq": 8.44, "rating": 28.17,
                        "experience": 8.60, "distance": 38.21 } } }
  ]
}
```

Validation: NVQ must be 1 to 7 and rating 0 to 5; an empty worker list returns 400; an out-of-range field returns 422; a category with no matching workers returns 200 with an empty list. GET /weights returns the active weights (NVQ 0.0844, rating 0.3045, experience 0.2151, distance 0.3960), criterion directions, normalisation bounds (NVQ

3–7, rating 1–5, experience 0–25 years, distance 0–50 km) and derivation metadata. GET /health reports readiness, and POST /score scores a single worker without filtering.

A.3 Component 3: review credibility and trust score

The C3 service exposes two operations.

Table A.1: Component 3 operations and fields

Operation	Request fields	Response fields
Analyse a review	Worker identifier; service category (39 labels); star rating (1–5); optional review text	Predicted label (Negative, Neutral, Positive); confidence; sentiment value (−1 to +1); updated category score
Get a trust score	Worker identifier; service category	Final trust score (0–10); number of reviews; credibility factor C = N/(N + 5)

A.4 Component 4: repair price estimator

```
POST /api/estimate-repair
{ "trade": "Plumbing", "task": "Pipe Leak Repair", "sizing": "Moderate" }

200 OK
{ "trade": "Plumbing", "task": "Pipe Leak Repair", "sizing": "Moderate",
  "price_min": 2500, "price_max": 4200, "currency": "LKR" }
```

A.5 Component 4: transport fee engine

```
POST /api/estimate-transport
{ "origin": { "lat": 6.9344, "lng": 79.8428 },
  "destination": { "lat": 6.8883, "lng": 79.8588 } }

200 OK
{ "distanceKm": 5.4, "trafficDelayMins": 7, "weatherCondition": "Rain",
  "totalTransportFee": 1109,
  "breakdown": { "baseFare": 200.0, "distanceCost": 511.06,
                 "trafficMultiplier": 1.30, "weatherMultiplier": 1.20,
                 "fuelRatio": 1.183 } }
```

The fee is computed as round([200 + 5.4 × 80 × 1.183] × 1.30 × 1.20) = LKR 1,109. Traffic multipliers are 1.00 for a delay of up to 15%, 1.15 for 15–40% and 1.30 above 40%; weather multipliers are 1.00 for clear conditions, 1.20 for rain or drizzle and 1.35 for thunderstorms.

Appendix B Team contribution declaration

The members of group R26-IT-090 confirm the following record of individual and shared contributions. Detailed individual contributions are stated in Section 1.8 of each individual thesis.

Table B.1: Record of individual and shared contributions

Member	Individual contribution	Shared contribution
Bandara W A M S (IT22551320)	Component 1: 39-category taxonomy, problem description dataset, FAISS and SVM models, evaluation, intent recognition service, worker_type contract	Flutter application, Node.js backend and database schema; integration; end-to-end testing; this report
Shiwanthika M M (IT22283658)	Component 2: survey instrument and data collection, regression and weight derivation, ranking engine and service, functional and performance testing	Flutter application, Node.js backend and database schema; integration; end-to-end testing; this report
Wijebandara W.N.S.K (IT22643704)	Component 3: augmented multilingual sentiment dataset, XLM-R fine-tuning, trust-scoring model, review analysis service	Flutter application, Node.js backend and database schema; integration; end-to-end testing; this report
Kumara A.J.T.M (IT22137982)	Component 4: transport fee model and service, repair price survey and dataset, decision tree and random forest models, estimator service, automated tests and deployment	Flutter application, Node.js backend and database schema; integration; end-to-end testing; this report

Appendix C Shared worker category list

Table C.1 lists the 39 worker categories shared by all components. The machine label is produced by Component 1 and used in API messages; the readable name is shown in the application and used in Component 4's repair price dataset.

Table C.1: The 39 shared worker categories

#	Machine label (worker_type)	Readable name	#	Machine label (worker_type)	Readable name
1	ac_technician	AC technician	21	painter	Painter
2	aluminium_fabricator	Aluminium fabricator	22	pest_control_specialist	Pest control specialist
3	bathroom_water_heater	Bathroom water heater	23	plumber	Plumber
4	carpenter	Carpenter	24	refrigerator_technician	Refrigerator technician
5	cctv_installer	CCTV installer	25	rice_cooker_technician	Rice cooker technician

#	Machine label (worker_type)	Readable name	#	Machine label (worker_type)	Readable name
6	ceiling_installer	Ceiling installer	26	roof_repairer	Roof repairer
7	curtain_blind_specialist	Curtain blind specialist	27	septic_tank_service	Septic tank service
8	drain_unblocking	Drain unblocking	28	sewing_machine_technician	Sewing machine technician
9	electric_oven_technician	Electric oven technician	29	solar_panel_technician	Solar panel technician
10	electrician	Electrician	30	tiler	Tiler
11	electronics_technician	Electronics technician	31	tree_service_specialist	Tree service specialist
12	fan_technician	Fan technician	32	upholstery_specialist	Upholstery specialist
13	furniture_restoration	Furniture restoration	33	washing_machine_technician	Washing machine technician
14	gas_cooker_technician	Gas cooker technician	34	washroom_cleaner	Washroom cleaner
15	glass_installer	Glass installer	35	water_pump_technician	Water pump technician
16	house_cleaner	House cleaner	36	water_tank_cleaner	Water tank cleaner
17	kitchen_appliance_technician	Kitchen appliance technician	37	waterproofing_specialist	Waterproofing specialist
18	kitchen_water_heater	Kitchen water heater	38	welder	Welder
19	lawn_care_worker	Lawn care worker	39	well_cleaner	Well cleaner
20	mason	Mason			

Appendix D Academic integrity and AI use disclosure

Generative AI tools were used during this project in the capacities listed in Table D.1. The team remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions of this report. Each individual thesis contains its own disclosure for component-level work.

Table D.1: Disclosure of AI tool use in the project and in this report

Tool	Purpose	How it was used	Verification performed
Claude (Anthropic)	Generating part of Component 1's training and test descriptions	Prompted with team-written seed phrases and problems re-written from public Facebook posts	Duplicates and ambiguous rows removed; limitations disclosed in the C1 thesis

Tool	Purpose	How it was used	Verification performed
AI assistants	Drafting and reviewing code for component services, analysis scripts and tests	Suggestions reviewed, executed and checked against independently computed results	All reported figures produced by executing code on the collected data
Claude (Anthropic)	Structuring and drafting this common report from the four individual theses, the template and the team's answers	Drafted chapters, tables and diagrams from the theses' reported results	The team reviewed every section and checked each number against the individual theses